MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE AMH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic AND-OR-INVERT gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of procedure. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.239 failures per million operating hours for class I or 2.3634 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be AMH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5 4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/252-1 for class I and M28787/252-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3.0 to 18.0 V $\pm$5 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through TT shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows. A, B, C where applicable, followed by the remaining tests in the sequence.

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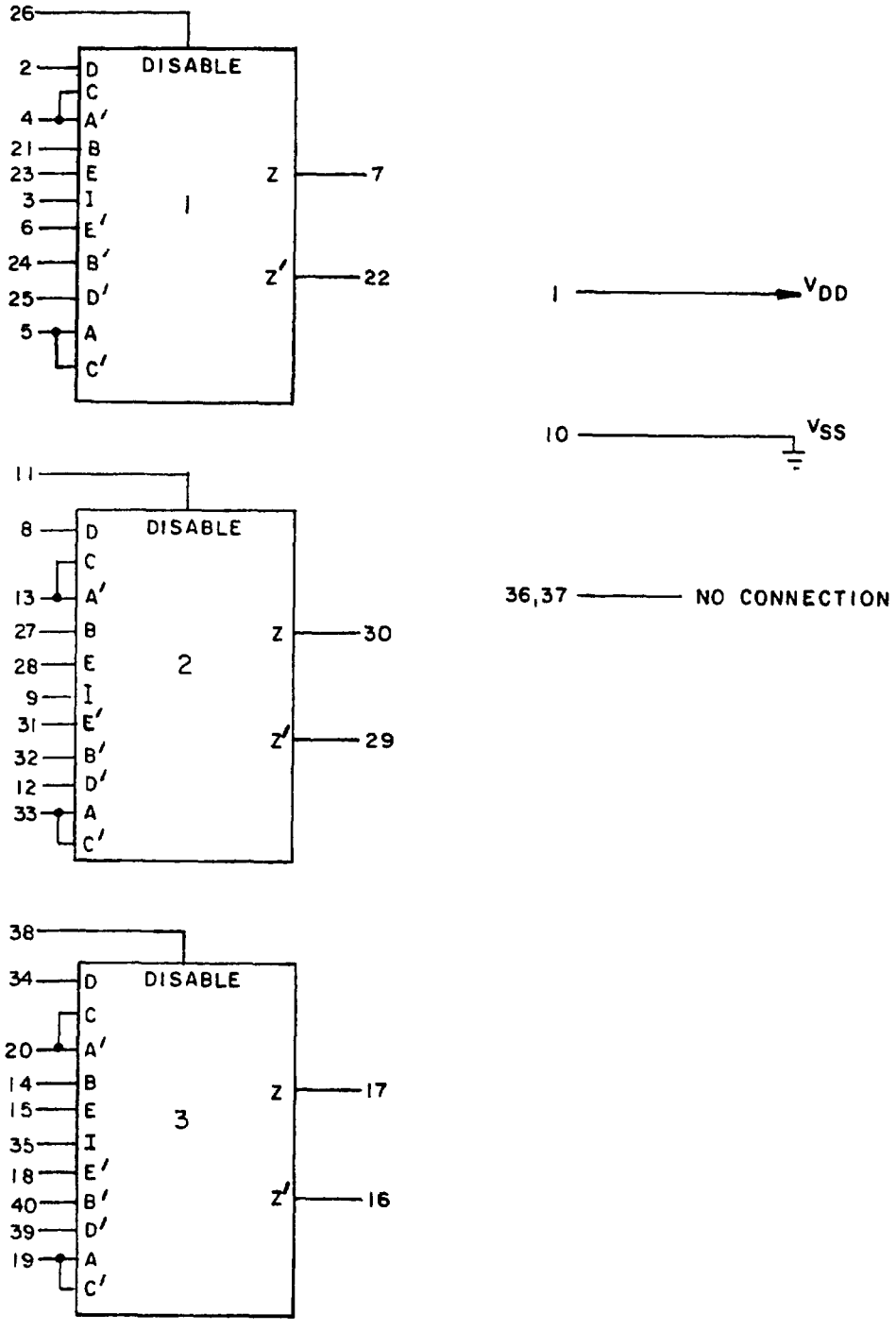


FIGURE 1 AMH module function diagram, AND-OR-INVERT gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B30	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C30	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D6	Output leakage current, high level, V_{DD} = 15 V	VI	X	-0.10	0.10	-0.10	0.10	-1.45	1.45	-0.10	0.10	-3.0	3.0	µA
E1-E6	Output leakage current, low level, V_{DD} = 15 V	VII	X	-0.10	0.10	-0.10	0.10	-1.45	1.45	-0.10	0.10	-3.0	3.0	µA
F1-F6	Static output voltage, high level, V_{DD} = 5 V (I_{OH} = 0)	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G6	Static output voltage, low level, V_{DD} = 5 V (I_{OL} = 0)	IX	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
H1-H6	Static output voltage, high level, V_{DD} = 5 V	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
J1-J6	Static output voltage, low level, V_{DD} = 5 V	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
K1-K6	Static output voltage, high level, V_{DD} = 15 V	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
L1-L6	Static output voltage, low level, V_{DD} = 15 V	XIII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
M1-M64	Functional test	XIV	X	← See table XIV →										
N1-N30	Input current, high level	XV	X	0	200	0	200	0	200	---	---	0	200	nA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P30	Input current, low level	XVI	X	0	-200	0	-200	0	-200	---	---	0	-200	nA
Q1	Static power supply current, low level	XVII	X	0	12	0	12	0	175	0	12	0	360	µA
Q2	Static power supply current, high level	XVII	X	0	12	0	12	0	175	0	12	0	360	µA
R1-R6	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	35	350	35	350	38	385	26	265	43	430	ns
R7-R12	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	83	830	83	830	91	915	63	630	101	1015	ns
S1-S6	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XIX	S	17	175	17	175	19	195	13	135	21	215	ns
S7-S12	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XIX	Q	65	655	65	655	72	725	50	500	87	875	ns
T1-T24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XX	S	44	440	44	440	48	485	33	335	54	540	ns
T25-T30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XX	S	27	270	27	270	30	300	20	205	33	330	ns
T31-T36	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XX	S	36	360	36	360	40	400	27	275	44	440	ns
T37-T60	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	64	640	64	640	70	705	48	485	78	785	ns
T61-T66	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	47	470	47	470	52	520	35	355	57	575	ns
T67-T72	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XX	Q	56	560	56	560	62	620	42	425	68	685	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	36	360	36	360	40	400	27	275	44	440	ns
U25-U30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	27	270	27	270	30	300	20	205	33	330	ns
U31-U36	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XXI	S	36	360	36	360	40	400	27	275	44	440	ns
U37-U60	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	56	560	56	560	62	620	42	425	68	685	ns
U61-U66	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	47	470	47	470	52	520	35	355	57	575	ns
U67-U72	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XXI	Q	56	560	56	560	62	620	42	425	68	685	ns
V1-V6	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 5 V	XXII	S	25	250	25	250	27	275	19	190	30	305	ns
V7-V12	Disable delay time (t_{PZH}), C_L = 150 pF, V_{DD} = 5 V	XXII	Q	45	450	45	450	50	500	34	340	55	550	ns
W1-W6	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 5 V	XXIII	S	25	250	25	250	26	265	22	220	28	280	ns
W7-W12	Disable delay time (t_{PHZ}), C_L = 150 pF, V_{DD} = 5 V	XXIII	Q	45	450	45	450	48	485	37	370	52	525	ns
X1-X6	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 5 V	XXIV	S	13	135	13	135	15	150	10	100	16	165	ns
X7-X12	Disable delay time (t_{PZL}), C_L = 150 pF, V_{DD} = 5 V	XXIV	Q	33	335	33	335	37	370	25	255	41	410	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y6	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 5 V	XXV	S	25	255	25	255	28	280	19	195	31	315	ns
Y7-Y12	Disable delay time (t_{PLZ}), C_L = 150 pF, V_{DD} = 5 V	XXV	Q	45	455	45	455	50	505	34	345	55	555	ns
Z1-Z6	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XXVI	Q	15	150	15	150	16	165	11	115	18	185	ns
AA1-AA6	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XXVII	Q	7	75	7	75	8	83	5	57	9	92	ns
BB1-BB24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	16	165	16	165	18	185	12	125	20	200	ns
BB25-BB30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	11	110	11	110	12	120	8	85	13	135	ns
BB31-BB36	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXVIII	Q	15	150	15	150	16	165	11	115	18	185	ns
CC1-CC24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIX	Q	13	135	13	135	15	150	10	105	16	165	ns
CC25-CC30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIX	Q	9	90	9	90	9	99	6	68	11	110	ns
CC31-CC36	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXIX	Q	14	140	14	140	15	155	10	105	17	170	ns
DD1-DD6	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 10 V	XXX	Q	10	100	10	100	11	110	7	75	12	125	ns
EE1-EE6	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 10 V	XXXI	Q	10	100	10	100	11	110	8	80	12	120	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF1-FF6	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 10 V	XXXII	Q	10	100	10	100	11	110	8	80	12	120	ns
GG1-GG6	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 10 V	XXXIII	Q	10	105	10	105	11	115	8	80	13	130	ns
HH1-HH6	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXXIV	Q	11	110	11	110	12	120	8	85	13	135	ns
JJ1-JJ6	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXXV	Q	5	55	5	55	6	61	4	42	6	67	ns
KK1-KK24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	11	110	11	110	12	120	8	85	13	135	ns
KK25-KK30	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	7	75	7	75	8	83	5	57	9	92	ns
KK31-KK36	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXXVI	Q	10	100	10	100	11	110	7	75	12	125	ns
LL1-LL24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXVII	Q	11	110	11	110	12	120	8	85	13	135	ns
LL25-LL30	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXVII	Q	6	60	6	60	6	66	4	45	7	74	ns
LL31-LL36	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXXVII	Q	10	100	10	100	11	110	7	75	12	125	ns
MM1-MM6	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 15 V	XXXVIII	Q	8	80	8	80	8	86	6	65	9	95	ns
NN1-NN6	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 15 V	XXXIX	Q	8	80	8	80	8	86	6	65	9	95	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
PP1-PP6	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 15 V	XL	Q	8	80	8	80	8	86	6	65	9	95	ns
QQ1-QQ6	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 15 V	XLI	Q	8	80	8	80	8	86	6	65	9	95	ns
RR	Power supply filter capacitance	XLII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
SS1-SS24	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF
SS25-SS30	Input capacitance	XLIII	Q	4	25	4	25	4	25	4	25	4	25	pF
TT1-TT6	Output capacitance	XLIV	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through TT in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through M, Q through QQ, and RR through TT shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions N and P shall be performed at 25°, 60°, and 100°C only, using the initial limits for the applicable operating temperature.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition RR, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through PP shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through TT shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits for table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	5.4	5.4	mW
Input voltage			
Low level (V_{DD} = 5 V)	0 min, 1.5 max	0 min, 1.5 max	V
Low level (V_{DD} = 15 V)	0 min, 4.0 max	0 min, 4.0 max	V
High level (V_{DD} = 5 V)	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
High level (V_{DD} = 15 V)	11.0 min, V_{DD} max	11.0 min, V_{DD} max	V
Input current (V_{DD} = 18 V)			
Low level (V_{IL} = 0.0 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V)	0 min, 200 max	0 min, 200 max	nA

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	Class I	Class II	Unit
Output drive current (V_{DD} = 5V)			
Low level (V_{OL} = 0.4 V)			
I_{OL} at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	---	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4.6 V)			
I_{OH} at -55°C	---	0.25 min	mA
0° and 25°C	0.20 min	0.20 min	mA
60°C	0.17 min	0.17 min	mA
100°C	---	0.14 min	mA

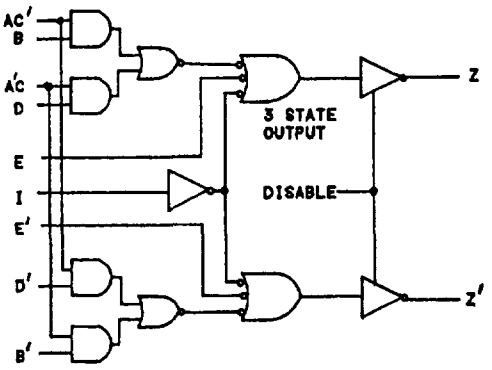
6.5 Module function. The module contains 3 independent, dual 2 wide, 2 input expandable AND-OR-INVERT gates with three state outputs.

6.6 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

A/A'	B	C/C'	D	E/E'	Inhibit	DISABLE	Z	Z'
L	L	L	L	H	L	L	H	H
L	X	L	X	H	L	L	H	H
L	X	X	L	H	L	L	H	H
X	L	L	X	H	L	L	H	H
X	L	X	L	H	L	L	H	H
H	H	X	X	X	X	L	L	L
X	X	H	H	X	X	L	L	L
X	X	X	X	L	X	L	L	L
X	X	X	X	X	H	L	L	L
X	X	X	X	X	X	H	*	*

1/ H = High voltage level.
2/ L = Low voltage level.
3/ X = Don't care.
4/ * = High impedance state.



$Z = \overline{(AB + CD + E + I)}$

$Z' = \overline{(A'B' + CD' + E' + I)} = \overline{(CB' + AD' + E' + I)}$

6.6.1 Paralleling of outputs. Paralleling of outputs is not permissible.

6.7 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed.

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.
b. Soldering-iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply is off.
d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.
e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the input protection diodes.
f. All unused input leads must be connected to either V_{SS} (ground) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

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6.8 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula

$$P_{Total} = (CV_{DD}^2 f)N + \text{quiescent power dissipation}$$

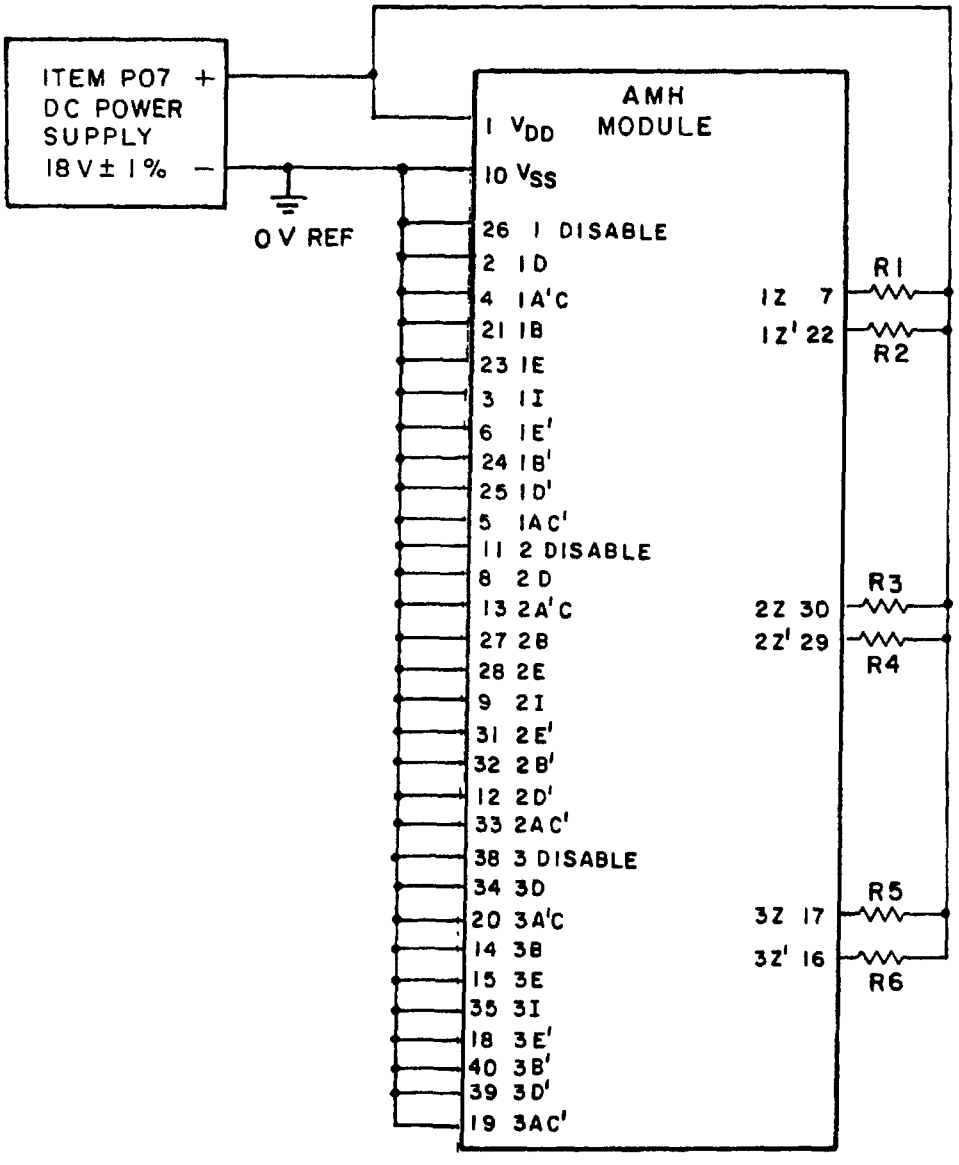
where. C = Load capacitance.
V_{DD} = Supply voltage.
f = Operating frequency.
N = Number of outputs.

Agent:
DLA - ES

Preparing activity.
Navy - EC

(Project 5963-N199)

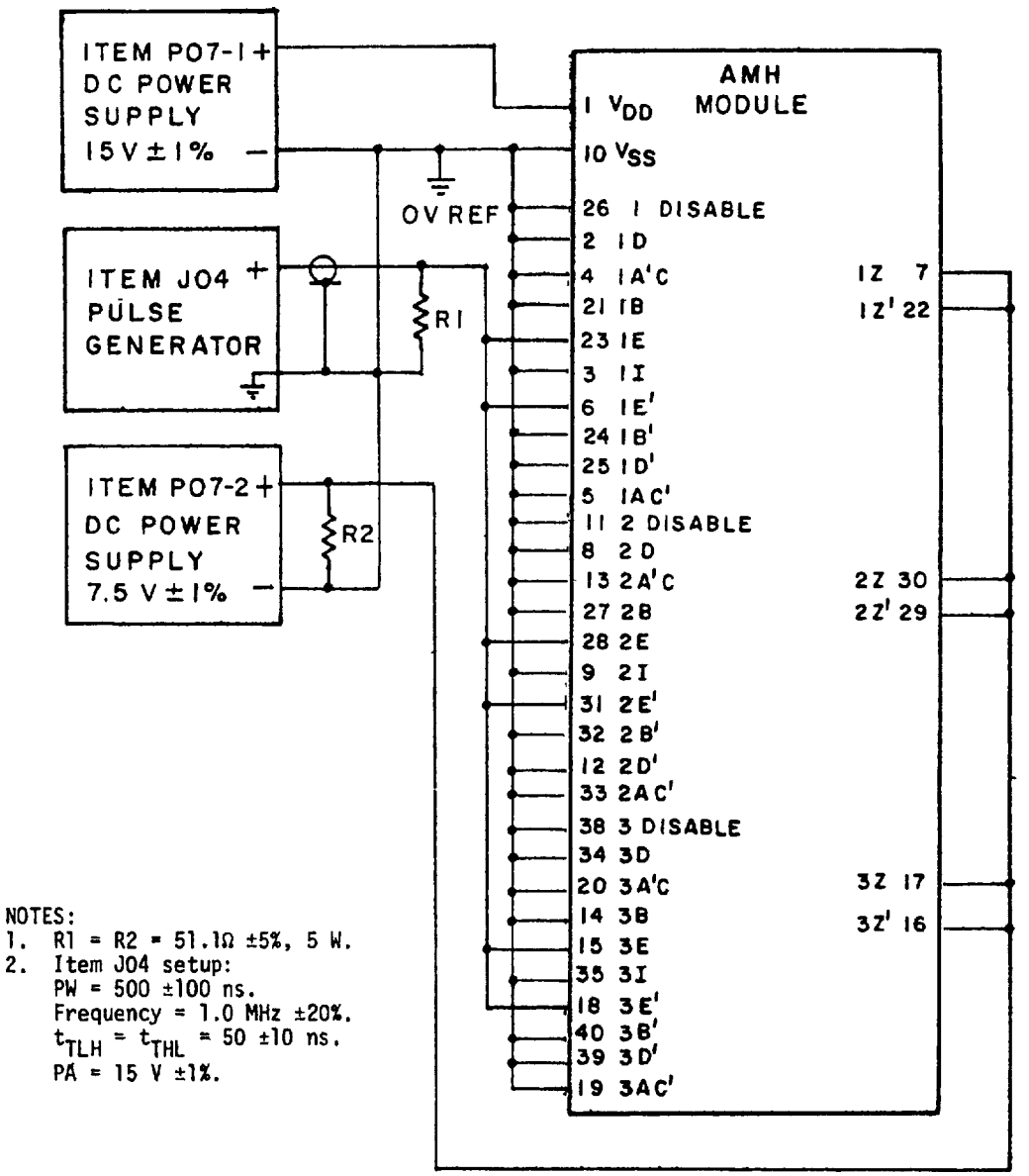
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NOTE R1-R6 = 6 81 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 5 W.
2. Item J04 setup:
   PW = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 15 V ±1%.

FIGURE 3 Component temperature test circuit.

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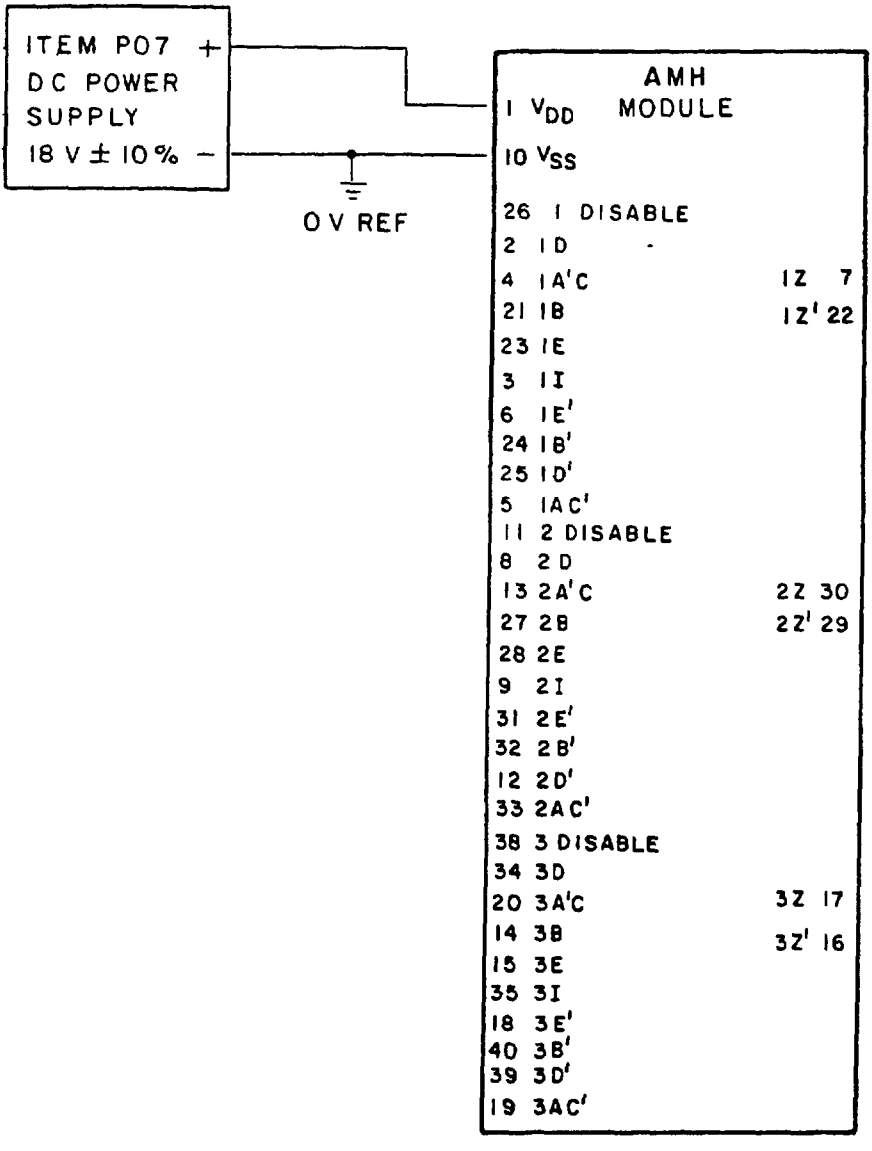


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	C, 36, 37	Frame	10	∞	MΩ	5
A2	Isolation, unused pin	C	36	10	∞	MΩ	5
A3	Isolation, unused pin	C	37	10	∞	MΩ	5

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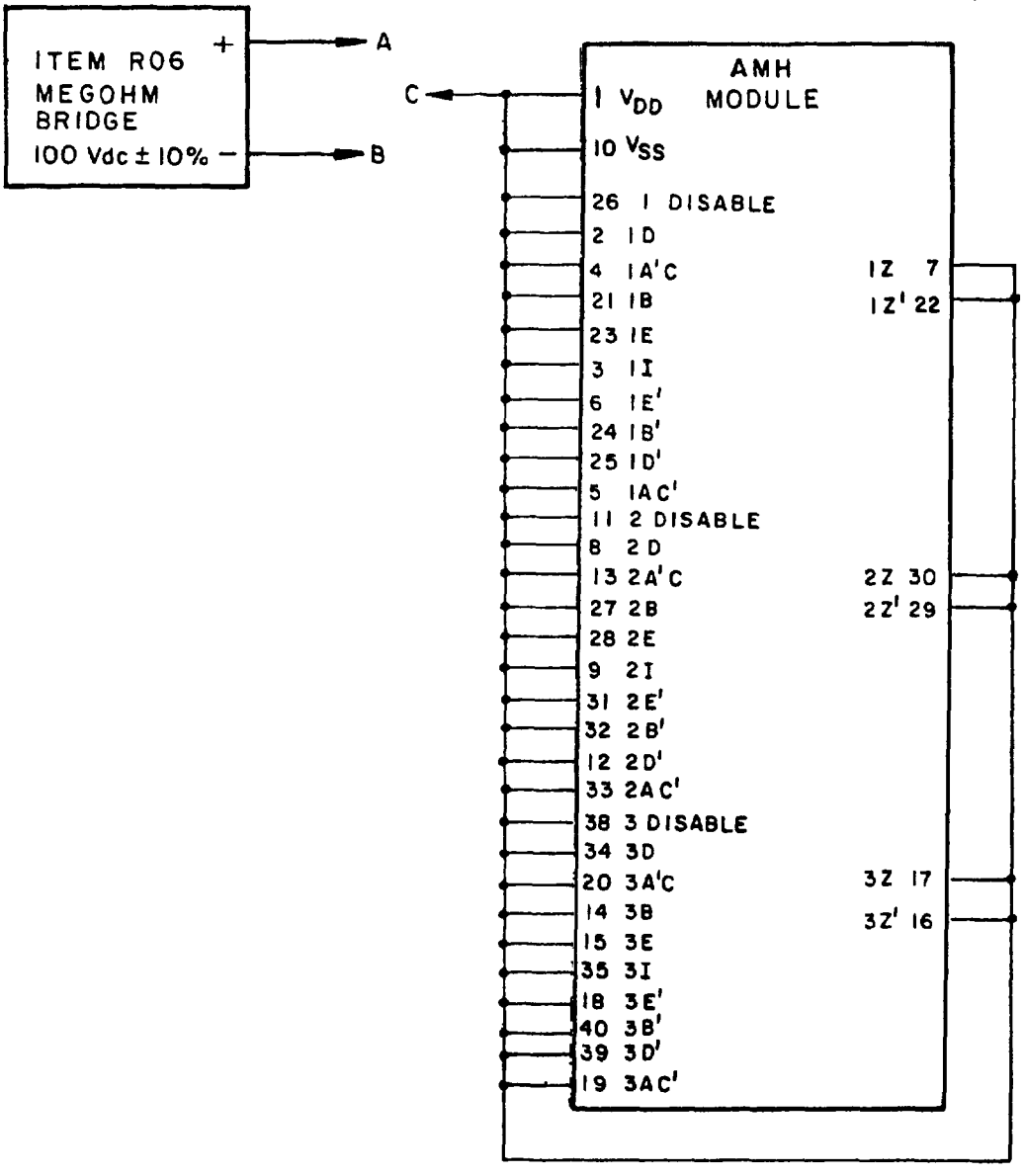


FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1 DISABLE Input clamp diode voltage, negative	26	Open	-0.4	-6.0	V	6
B2	1D	2					
B3	1A'C	4					
B4	1B	21					
B5	1E	23					
B6	1I	3					
B7	1E'	6					
B8	1B'	24					
B9	1D'	25					
B10	1AC'	5					
B11	2 DISABLE	11					
B12	2D	8					
B13	2A'C	13					
B14	2B	27					
B15	2E	28					
B16	2I	9					
B17	2E'	31					
B18	2B'	32					
B19	2D'	12					
B20	2AC'	33					
B21	3 DISABLE	38					
B22	3D	34					
B23	3A'C	20					
B24	3B	14					
B25	3E	15					
B26	3I	35					
B27	3E'	18					
B28	3B'	40					
B29	3D'	39					
B30	3AC'	19					

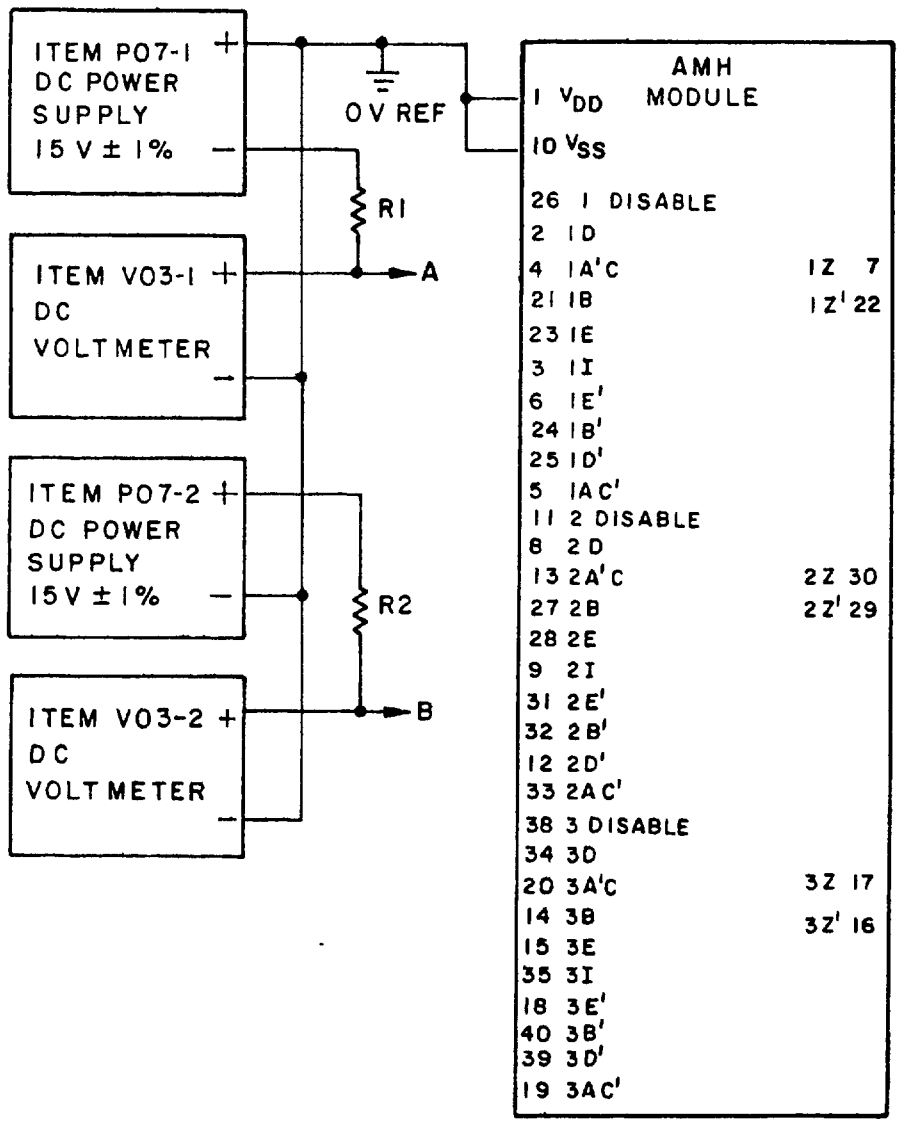
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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
C1	1 DISABLE	Open	26	0.4	1.5	V	6
C2	1D		2				
C3	1A'C		4				
C4	1B		21				
C5	1E		23				
C6	1I		3				
C7	1E'		6				
C8	1B'		24				
C9	1D'		25				
C10	1AC'		5				
C11	2 DISABLE		11				
C12	2D		8				
C13	2A'C		13				
C14	2B		27				
C15	2E		28				
C16	2I		9				
C17	2E'		31				
C18	2B'		32				
C19	2D'		12				
C20	2AC'		33				
C21	3 DISABLE		38				
C22	3D		34				
C23	3A'C		20				
C24	3B		14				
C25	3E		15				
C26	3I		35				
C27	3E'		18				
C28	3B'		40				
C29	3D'		39				
C30	3AC'		19				

Input clamp diode voltage, positive

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since dc power supplies PO7-1 and PO7-2 and dc voltmeters VO3-1 and VO3-2 are not used simultaneously, only one dc power supply, PO7 and one dc voltmeter VO3, is required

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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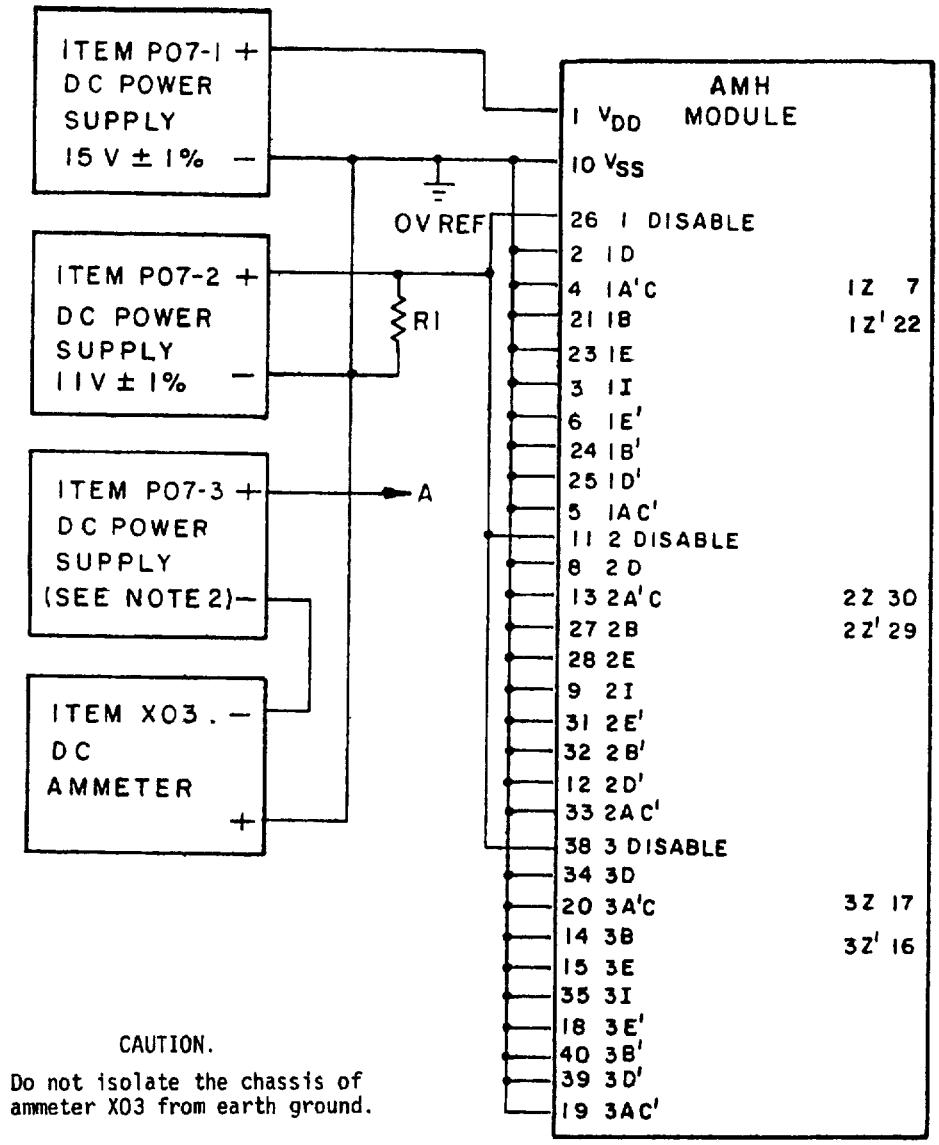
TABLE VI. Output leakage current (high-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1Z	Output leakage current, high level, V_{DD} = 15 V	7	-0.10	0.10	µA	7
D2	1Z'		22				
D3	2Z		30				
D4	2Z'		29				
D5	3Z		17				
D6	3Z'		16				

TABLE VII. Output leakage current (low-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1Z	Output leakage current, low level, V_{DD} = 15 V	7	-0.10	0.10	µA	7
E2	1Z'		22				
E3	2Z		30				
E4	2Z'		29				
E5	3Z		17				
E6	3Z'		16				

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NOTES:
1. R1 = 51.1Ω ±5%, 2 W.
2 Set item P07-3 at 13.5 V ±1% for test condition D and at 1.5 V ±1% for test condition E.

FIGURE 7. Output leakage current (high-level and low-level) test circuit for tables VI and VII.

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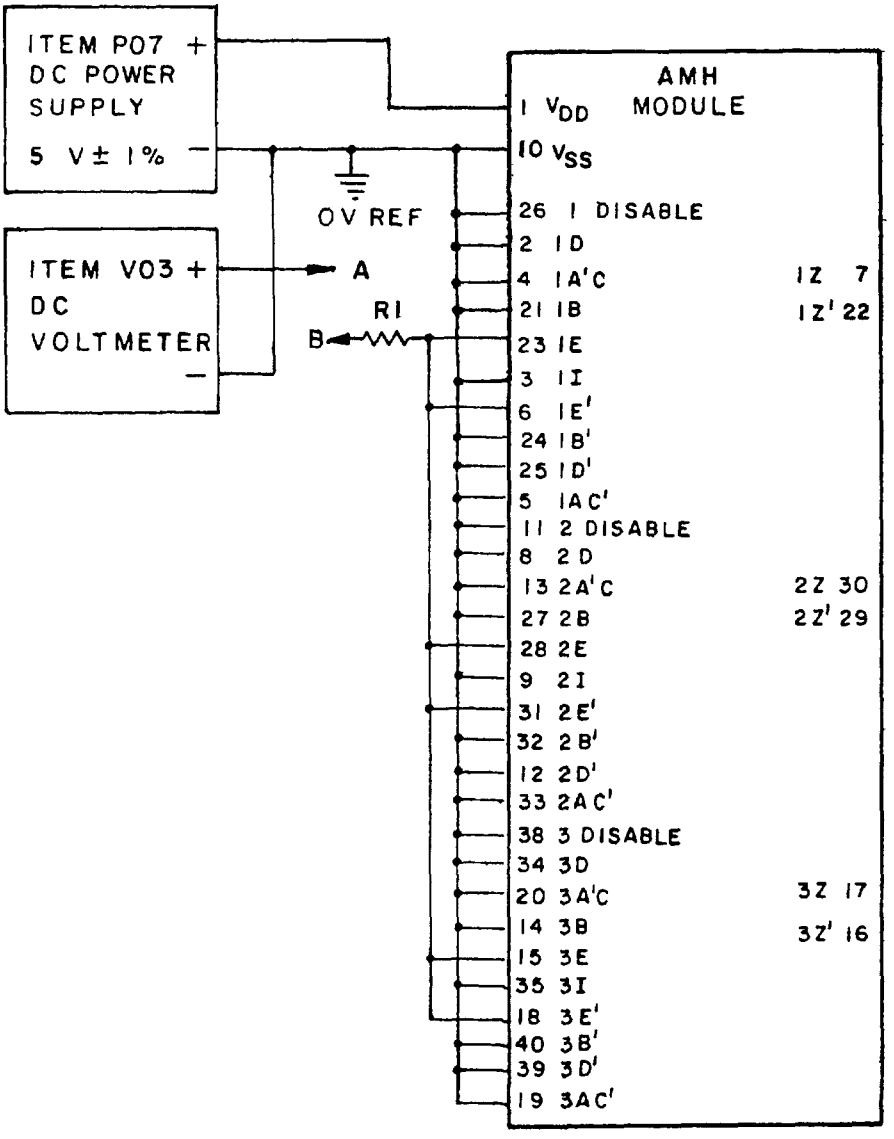
TABLE VIII. Static output voltge (high-level) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1Z	Static output voltage, high level, V_{DD} = 5 V, (I_{OH} = 0)	7	1	4.95	5.05	V	8
F2	1Z'		22					
F3	2Z		30					
F4	2Z'		29					
F5	3Z		17					
F6	3Z'		16					

TABLE IX. Static output voltge (low-level) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	1Z	Static output voltage, low level, V_{DD} = 5 V, (I_{OL} = 0)	7	10	0.00	0.05	V	8
G2	1Z'		22					
G3	2Z		30					
G4	2Z'		29					
G5	3Z		17					
G6	3Z'		16					

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NOTE R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 8 Static output voltage (high-level and low-level) test circuit for tables VIII and IX.

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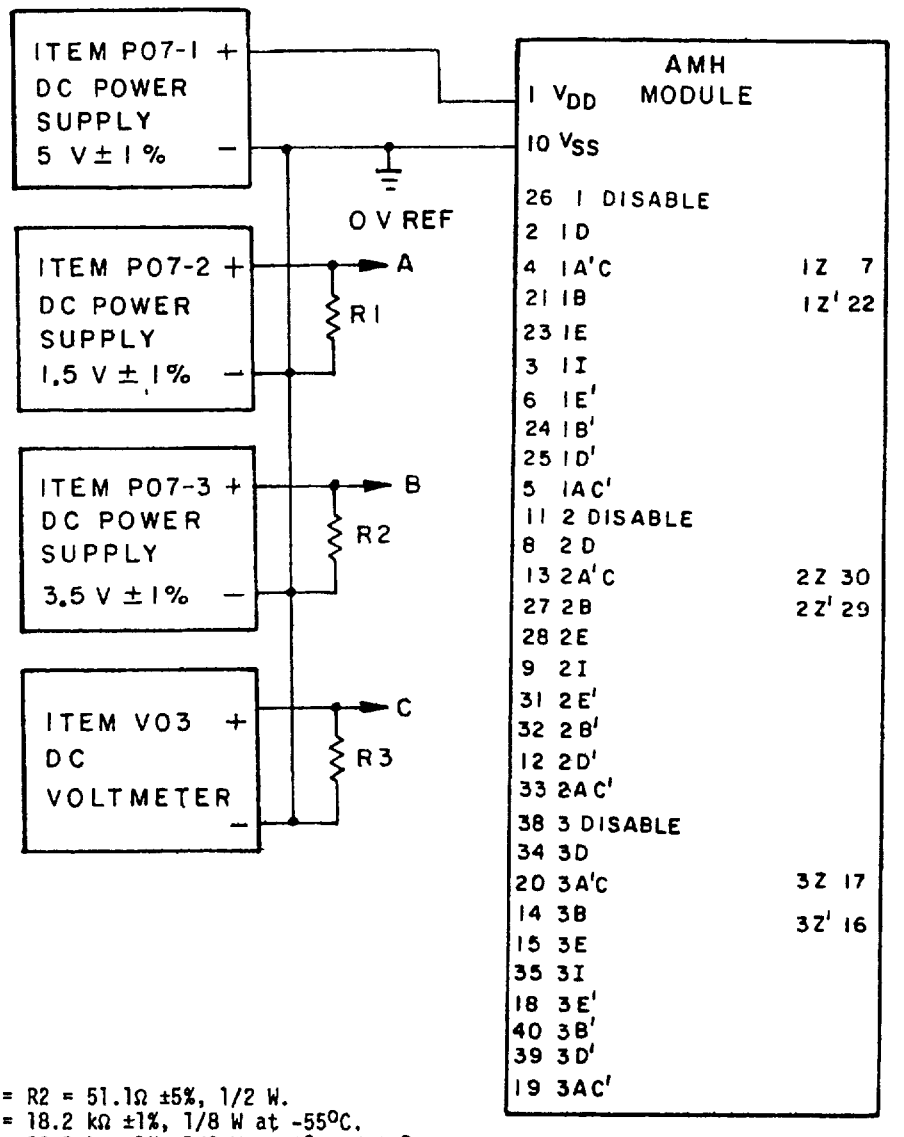
TABLE X. Static output voltage (high-level) test.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1Z	Static output voltage, high level, V_{DD} = 5 V	26,2,4,21,3,24, 25,5,11,8,13, 27,7,32,12,33, 38,34,20,14,35, 40,39,19	23,6,28,31, 15,16	7	4.6	5.0	V	9
H2	1Z'				22				
H3	2Z				30				
H4	2Z'				29				
H5	3Z				17				
H6	3Z'				16				

TABLE XI. Static output voltage (low-level) test.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	1Z	Static output voltage, low level, V_{DD} = 5 V	26,2,4,21,23,6, 24,25,5,11,8,13, 27,28,31,32,12, 33,38,34,20,14, 15,18,40,39,19	3,9,35	7	0.0	0.4	V	10
J2	1Z'				22				
J3	2Z				30				
J4	2Z'				29				
J5	3Z				17				
J6	3Z'				16				

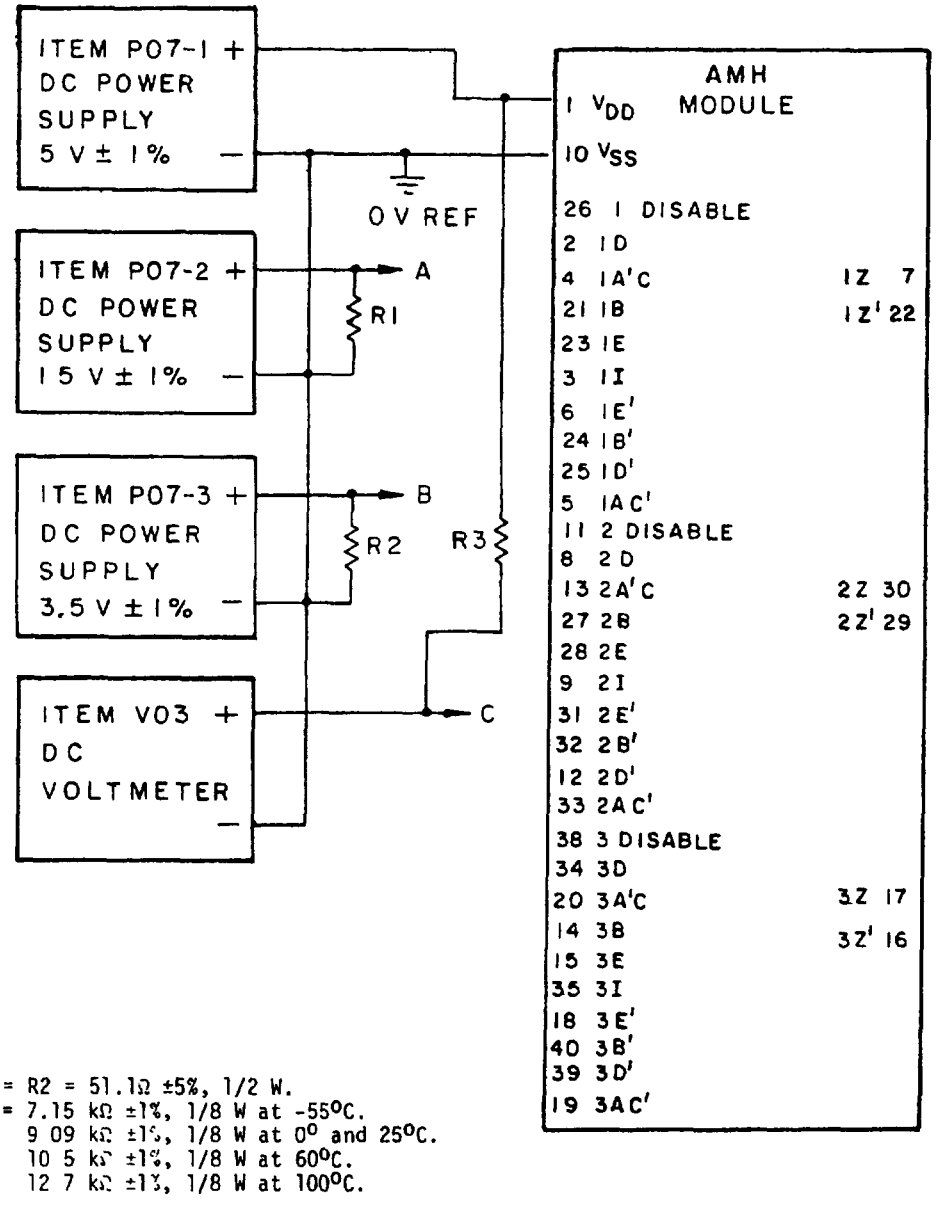
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/8 W at -55°C.
23.2 kΩ ±1%, 1/8 W at 0° and 25°C.
27.4 kΩ ±1%, 1/8 W at 60°C.
32.4 kΩ ±1%, 1/8 W at 100°C.

FIGURE 9. Static output voltage (high-level) test for table X.

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NOTES
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 7.15 kΩ ±1%, 1/8 W at -55°C.
9.09 kΩ ±1%, 1/8 W at 0° and 25°C.
10.5 kΩ ±1%, 1/8 W at 60°C.
12.7 kΩ ±1%, 1/8 W at 100°C.

FIGURE 10 Static output voltage (low-level) test for table XI.

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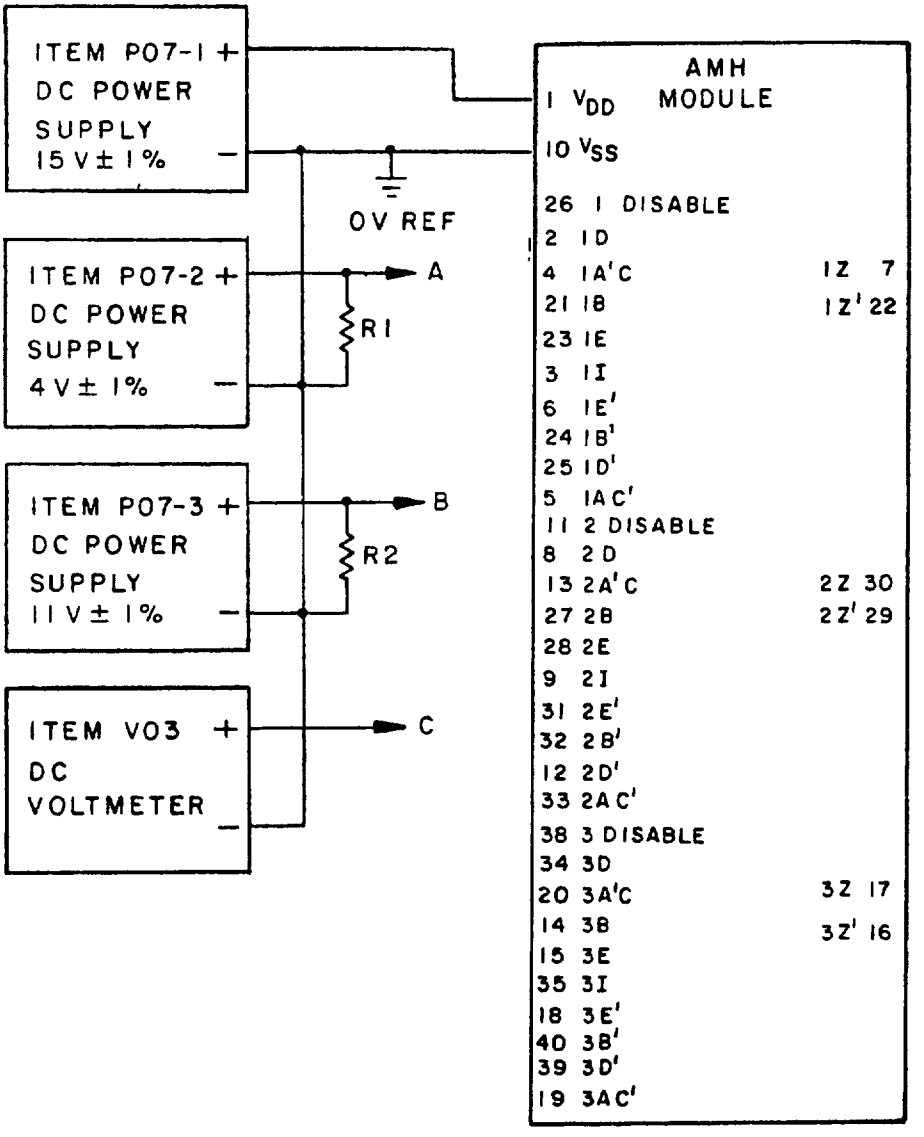
TABLE XII. Static output voltage (high-level) test.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	1Z	Static output voltage, high level, V_{DD} = 15 V	26,2,4,21,3,24, 25,5,11,8,13,27, 9,32,12,33,38, 34,20,14,35,40, 39,19	23,6,28,31, 15,18	7	13.5	15.0	V	11
K2	1Z'				22				
K3	2Z				30				
K4	2Z'				29				
K5	3Z				17				
K6	3Z'				16				

TABLE XIII. Static output voltage (low-level) test.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L1	1Z	Static output voltage, low level, V_{DD} = 15 V	26,2,4,21,23,6, 24,25,5,11,8,13, 27,28,31,32,12, 33,38,34,20,14, 15,18,40,39,19	3,9,35	7	0.0	1.5	V	11
L2	1Z'				22				
L3	2Z				30				
L4	2Z'				29				
L5	3Z				17				
L6	3Z'				16				

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NOTE: R1 = R2 = 51.1Ω ±5%, 5 W.

FIGURE 11 Static output voltage (high-level and low-level) test circuit for tables XII and XIII.

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TABLE XIV. Functional test.

Test condition	Parameter	Connect		C to output pin						Test figure
		A to pin	B to pin	7	22	30	29	17	16	
M1	Functional	26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	Open	L	L	L	L	L	L	12
M2		26,11,38,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	5,33,19	L	L	L	L	L	L	
M3		26,11,38,5,33,19,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	21,24,27,32,14,40	L	L	L	L	L	L	
M4		26,11,38,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	5,33,19,21,24,27,32,14,40,	L	L	L	L	L	L	
M5		26,11,38,5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	4,13,20	L	L	L	L	L	L	
M6		26,11,38,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	5,33,19,4,13,20	L	L	L	L	L	L	
M7		26,11,38,5,33,19,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	21,24,27,32,14,40,4,13,20	L	L	L	L	L	L	
M8		26,11,38,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	5,33,19,21,24,27,32,14,40,4,13,20	L	L	L	L	L	L	
M9		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,23,6,18,31,15,18,3,9,35	2,25,8,12,34,39	L	L	L	L	L	L	
M10		21,24,27,32,14,40,4,13,20,23,6,28,31,15,18,3,9,35,26,11,38	5,33,19,2,25,8,12,34,39	L	L	L	L	L	L	
M11		26,11,38,5,33,19,4,13,20,23,6,28,31,15,18,3,9,35	21,24,27,32,14,40,2,25,8,12,34,39	L	L	L	L	L	L	
M12		26,11,38,4,13,20,23,6,28,31,15,18,3,9,35	5,33,19,21,24,27,32,14,40,2,25,8,12,34,39	L	L	L	L	L	L	
M13		26,11,38,5,33,19,21,24,27,32,14,40,23,6,28,31,15,18,3,9,35	4,13,20,2,25,8,12,34,39	L	L	L	L	L	L	
M14		21,24,27,32,14,40,23,6,28,31,15,18,3,9,35,26,11,38	5,33,19,4,13,20,2,25,8,12,34,39	L	L	L	L	L	L	
M15		26,11,38,5,33,19,23,6,28,31,15,18,3,9,35	21,24,27,32,14,40,4,13,20,2,25,8,12,34,39	L	L	L	L	L	L	
M16		26,11,38,23,6,28,31,15,18,3,9,15	5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39	L	L	L	L	L	L	
M17		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,37,3,9,35	23,6,28,31,15,18	H	H	H	H	H	H	
M18		26,11,38,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,3,9,35	5,33,19,23,6,28,31,15,18	H	H	H	H	H	H	

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Connect		C to output pin						Test figure
		A to pin	B to pin	7	22	30	29	17	16	
M19	Functional	26,11,38,5,33,19,4,13,20,2,25,8,12,34,39,3,9,35	21,24,27,32,14,40,23,6,28,31,15,18	H	H	H	H	H	H	12
M20		26,11,38,4,13,20,2,25,8,12,34,39,3,9,35	5,33,19,21,24,27,32,14,40,23,6,28,31,15,18	L	H	L	H	L	H	
M21		26,11,38,5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,3,9,35	4,13,20,23,6,28,31,15,18	H	H	H	H	H	H	
M22		26,11,38,21,24,27,32,14,40,2,25,8,12,34,39,3,9,35	5,33,19,4,13,20,23,6,28,31,15,18	H	H	H	H	H	H	
M23		26,11,38,5,33,19,2,25,8,12,34,39,3,9,35	21,24,27,32,14,40,4,13,20,23,6,28,31,15,18	H	L	H	L	H	L	
M24		26,11,38,2,25,8,12,34,39,3,9,35	5,33,19,21,24,27,32,14,40,4,13,20,23,6,28,31,15,18	L	L	L	L	L	L	
M25		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,3,9,35	2,25,8,12,34,39,23,6,28,31,15,18	H	H	H	H	H	H	
M26		26,11,38,21,24,27,32,14,40,4,13,20,3,9,35	5,33,19,2,25,8,12,34,39,23,6,28,31,15,18	H	L	H	L	H	L	
M27		26,11,38,5,33,19,4,13,20,3,9,35	21,24,27,32,14,40 ?,25,8,12,34,39,23,6,28,31,15,18	H	H	H	H	H	H	
M28		26,11,38,4,13,20,3,9,35	5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18	L	L	L	L	L	L	
M29		26,11,38,5,33,19,21,24,27,32,14,40,3,9,35	4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	L	H	L	H	L	H	
M30		26,11,38,21,24,27,32,14,40,3,9,35	5,33,19,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	L	L	L	L	L	L	
M31		26,11,38,5,33,19,3,9,35	21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	L	L	L	L	L	L	
M32		26,11,38,3,9,35	5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	L	L	L	L	L	L	
M33		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	3,9,35	L	L	L	L	L	L	
M34		26,11,38,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	5,33,19,3,9,35	L	L	L	L	L	L	
M35		26,11,38,5,33,19,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	21,24,27,32,14,40,3,9,35	L	L	L	L	L	L	
M36		26,11,38,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18	5,33,19,21,24,27,32,14,40,3,9,35	L	L	L	L	L	L	
M37		26,11,38,5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18	4,13,20,3,9,35	L	L	L	L	L	L	

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TABLE XIV. Functional test - Continued.

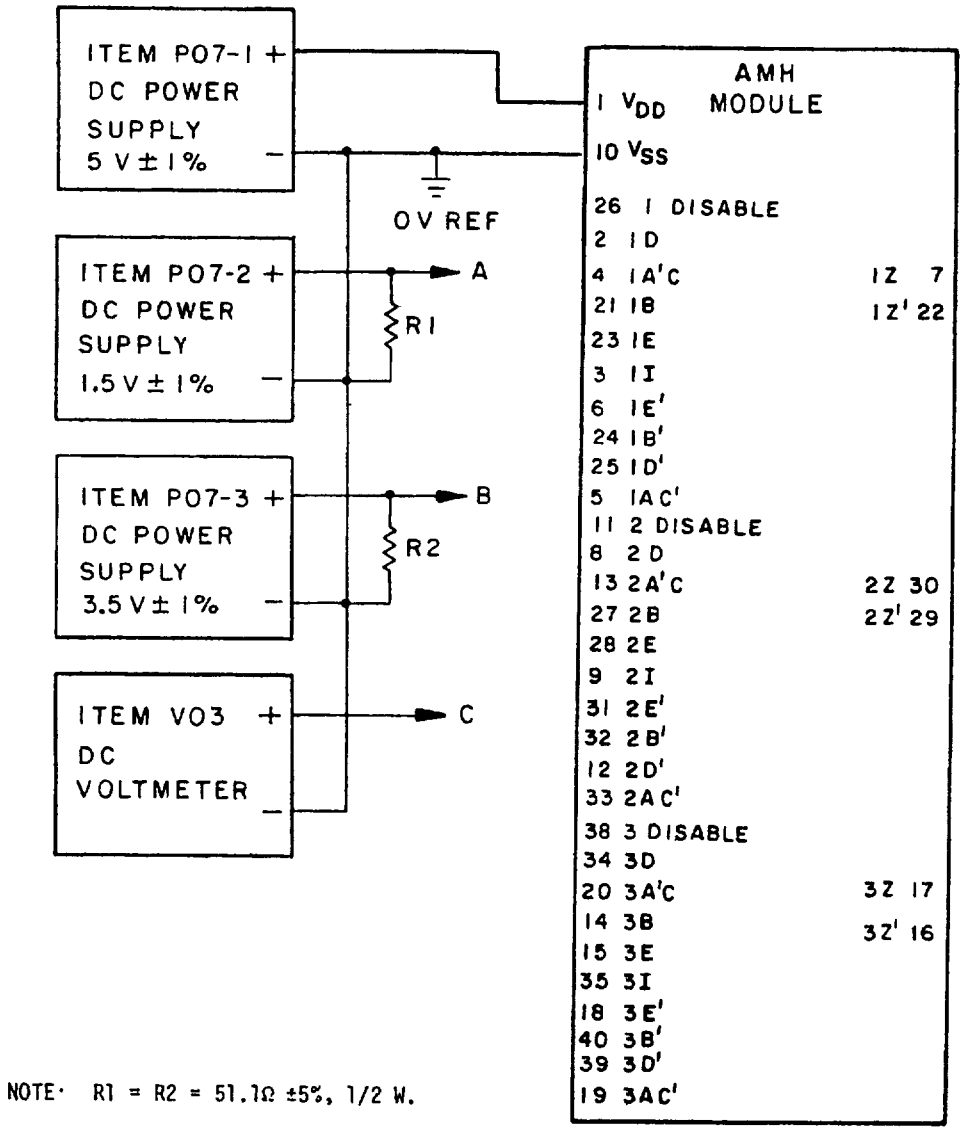
Test condition	Parameter	Connect		C to output pin						Test figure
		A to pin	B to pin	7	22	30	29	17	16	
M38	Functional	26,11,38,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18	5,33,19,4,13,20,3,9,35	L	L	L	L	L	L	12
M39		26,11,38,5,33,19,2,25,8,12,34,39,23,6,28,31,15,18	21,24,27,32,14,40,4,13,20,3,9,35	L	L	L	L	L	L	
M40		26,11,38,2,25,8,12,34,39,23,6,28,31,15,18	5,33,19,21,24,27,32,14,40,4,13,20,3,9,35	L	L	L	L	L	L	
M41		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,23,6,28,31,15,18	2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M42		26,11,38,21,24,27,32,14,40,4,13,20,23,6,28,31,15,18	5,33,19,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M43		26,11,38,5,33,19,4,13,20,23,6,28,31,15,18	21,24,27,32,14,40,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M44		26,11,38,4,13,20,23,6,28,31,15,18	5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M45		26,11,38,5,33,19,21,24,27,32,14,40,23,6,28,31,15,18	4,13,20,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M46		26,11,38,21,24,27,32,14,40,23,6,28,31,15,18	5,33,19,4,13,20,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M47		26,11,38,5,33,19,23,6,28,31,15,18	21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M48		26,11,38,23,6,28,31,15,18	5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,3,9,35	L	L	L	L	L	L	
M49		26,11,38,5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39	23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M50		26,11,38,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39	5,33,19,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M51		26,11,38,5,33,19,4,13,20,2,25,8,12,34,39	21,24,27,32,14,40,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M52		26,11,38,4,13,20,2,25,8,12,34,39	5,33,19,21,24,27,32,14,40,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M53		26,11,38,5,33,19,21,24,27,32,14,40,2,25,8,12,34,39	4,13,20,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M54		26,11,38,21,24,27,32,14,40,2,25,8,12,34,39	5,33,19,4,13,20,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M55		26,11,38,5,33,19,2,25,8,12,34,39	21,24,27,32,14,40,4,13,20,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M56		26,11,38,2,25,8,12,34,39	5,33,19,21,24,27,32,14,40,4,13,20,23,6,28,31,15,16,3,9,35	L	L	L	L	L	L	

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TABLE XIV. Functional test - Continued.

Test condition	Parameter	Connect		C to output pin						Test figure
		A to pin	B to pin	7	22	30	29	17	16	
M57	Functional	26,11,38,5,33,19,21,24,27,32,14,40,4,13,20	2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	12
M58		26,11,38,21,24,27,32,14,40,4,13,20	5,33,19,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M59		26,11,38,5,33,19,4,13,20	21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M60		26,11,38,4,13,20	5,33,19,21,24,27,32,14,40,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M61		26,11,38,5,33,19,21,24,27,32,14,40	4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M62		26,11,38,21,24,27,32,14,40	5,33,19,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M63		26,11,38,5,33,19	21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	
M64		26,11,38	5,33,19,21,24,27,32,14,40,4,13,20,2,25,8,12,34,39,23,6,28,31,15,18,3,9,35	L	L	L	L	L	L	

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NOTE: R1 = R2 = 51.1Ω ±5%, 1/2 W.

FIGURE 12. Functional test circuit for table XIV.

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TABLE XV. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	1 DISABLE Input current, high level	26	2,4,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19	0	200	nA	13
N2	1D	2	26,4,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N3	1A'C	4	26,2,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N4	1B	21	26,2,4,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N5	1E	23	26,2,4,21,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N6	1I	3	26,2,4,21,23,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N7	1E'	6	26,2,4,21,23,3,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N8	1B'	24	26,2,4,21,23,3,6,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N9	1D'	25	26,2,4,21,23,3,6,24,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N10	1AC'	5	26,2,4,21,23,3,6,24,25,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:04Z
Check the source to verify that this is the current version before use.

TABLE XV. Input current (high-level) test - Continued.

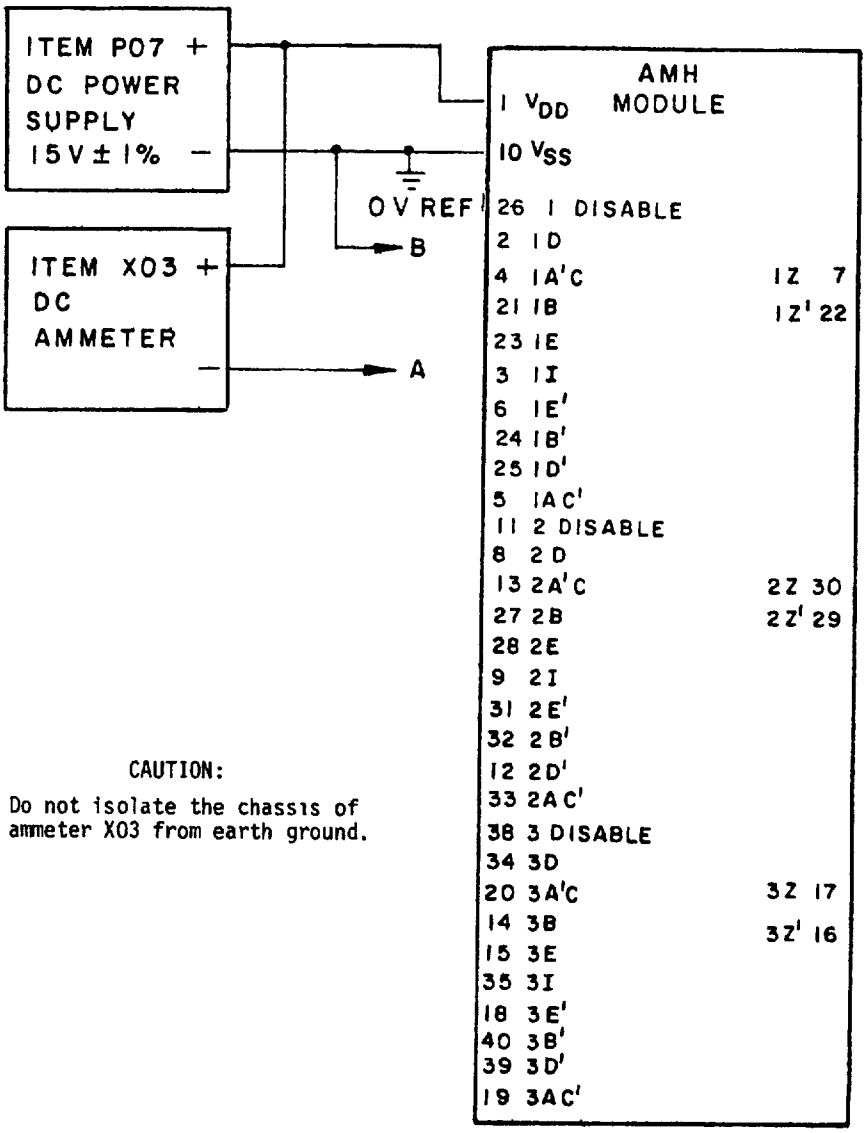
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N11	2 DISABLE	Input current, high level	11	26,2,4,21,23,3,6,24,25,5,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40.39,19	0	200	nA	13
N12	2D		8	26,2,4,21,23,3,6,24,25,5,11, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,40,39,19				
N13	2A'C		13	26,2,4,21,23,3,6,24,25,5,11,8, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N14	2B		27	26,2,4,21,23,3,6,24,25,5,11,8, 13,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N15	2E		28	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N16	2I		9	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N17	2E'		31	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,32,12,33,38,34,20, 14,15,35,18,40,39,19				
N18	2B'		32	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,12,33,38,34,20, 14,15,35,18,40,39,19				
N19	2D'		12	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,33,38,34,20, 14,15,35,18,40,39,19				
N20	3AC'		33	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,38,34,20, 14,15,35,18,40,39,19				

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TABLE XV. Input current (high-level) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Initial limits 25° ±5°C Ambient: Min	Initial limits 25° ±5°C Ambient: Max	Unit	Test figure
N21	3 DISABLE	Input current, high level	38	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,34,20, 14,15,35,18,40,39,19	0	200	nA	13
N22	3D		34	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,20, 14,15,35,18,40,39,19				
N23	3A'C		20	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 14,15,35,18,40,39,19				
N24	3B		14	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,15,35,18,40,39,19				
N25	3E		15	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,35,18,40,39,19				
N26	3I		35	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,18,40,39,19				
N27	3E'		18	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,40,39,19				
N28	3B'		40	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,39,19				
N29	3D'		39	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,40,19				
N30	3AC'		19	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,40,39				

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CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XV.

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TABLE XVI. Input current (low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ±5°C Ambient: Min	Initial limits 25° ±5°C Ambient: Max	Unit	Test figure
P1	1 DISABLE Input current, low level	26	2,4,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19	0	-200	nA	14
P2	1D	2	26,4,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P3	1A'C	4	26,2,21,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P4	1B	21	26,2,4,23,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P5	1E	23	26,2,4,21,3,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P6	1I	3	26,2,4,21,23,6,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P7	1E'	6	26,2,4,21,23,3,24,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P8	1B'	24	26,2,4,21,23,3,6,25,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P9	1D'	25	26,2,4,21,23,3,6,24,5,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P10	1AC'	5	26,2,4,21,23,3,6,24,25,11,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:04Z
Check the source to verify that this is the current version before use.

TABLE XVI. Input current (low-level) test - Continued.

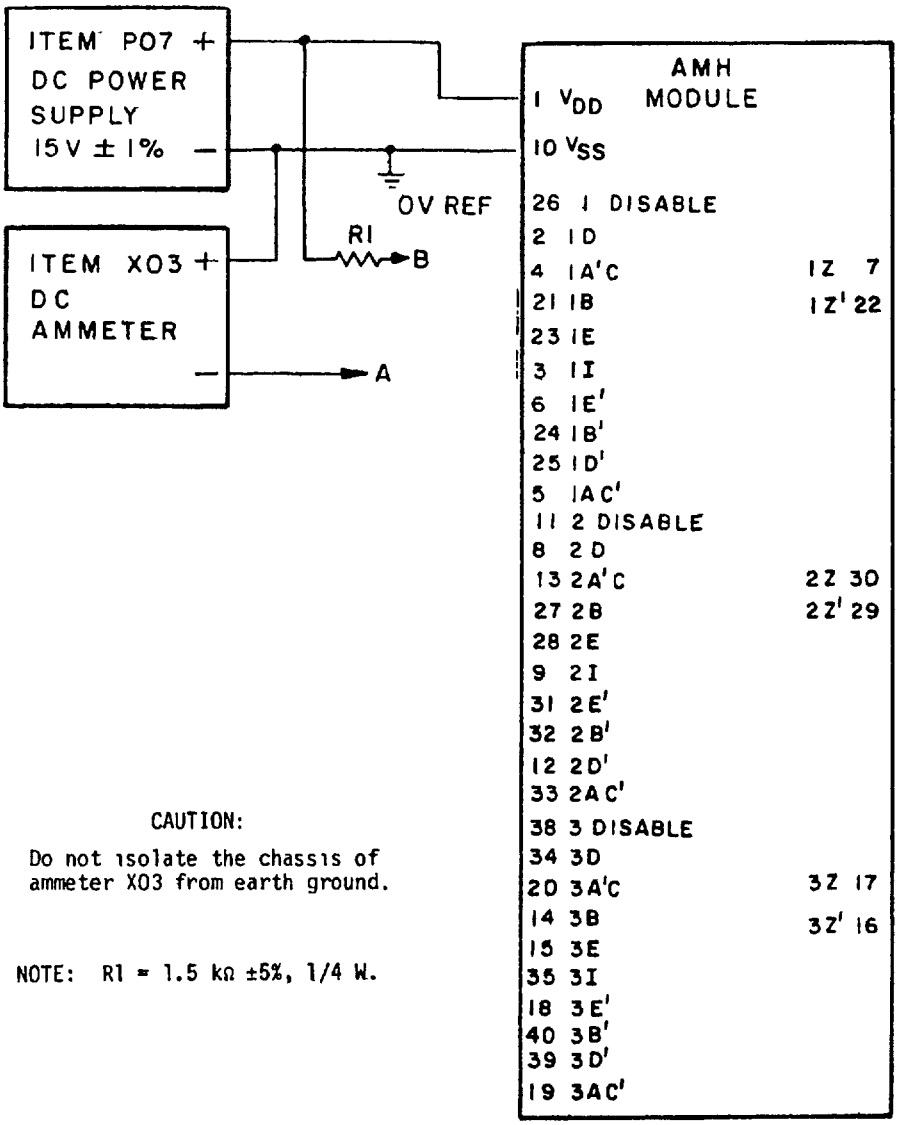
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P11	2 DISABLE Input current, low level	11	26,2,4,21,23,3,6,24,25,5,8,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19	0	-200	nA	14
P12	2D	8	26,2,4,21,23,3,6,24,25,5,11,13, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P13	2A'C	13	26,2,4,21,23,3,6,24,25,5,11,8, 27,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P14	2B	27	26,2,4,21,23,3,6,24,25,5,11,8, 13,28,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P15	2E	28	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,9,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P16	2I	9	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,31,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P17	2E'	31	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,32,12,33,38,34,20, 14,15,35,18,40,39,19				
P18	2B'	32	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,12,33,38,34,20, 14,15,35,18,40,39,19				
P19	2D'	12	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,33,38,34,20, 14,15,35,18,40,39,19				
P20	3AC'	33	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,38,34,20, 14,15,35,18,40,39,19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:04Z
Check the source to verify that this is the current version before use.

TABLE XVI. Input current (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
P21	3 DISABLE	Input current, low level	38	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,34,20, 14,15,35,18,40,39,19	0	-200	nA	14
P22	3D		34	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,20, 14,15,35,18,40,39,19				
P23	3A'C		20	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 14,15,35,18,40,39,19				
P24	3B		14	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,15,35,18,40,39,19				
P25	3E		15	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,35,18,40,39,19				
P26	3I		35	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,18,40,39,19				
P27	3E'		18	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,40,39,19				
P28	3B'		40	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,39,19				
P29	3D'		39	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,40,19				
P30	3AC'		19	26,2,4,21,23,3,6,24,25,5,11,8, 13,27,28,9,31,32,12,33,38,34, 20,14,15,35,18,40,39				

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CAUTION:

Do not isolate the chassis of ammeter XO3 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

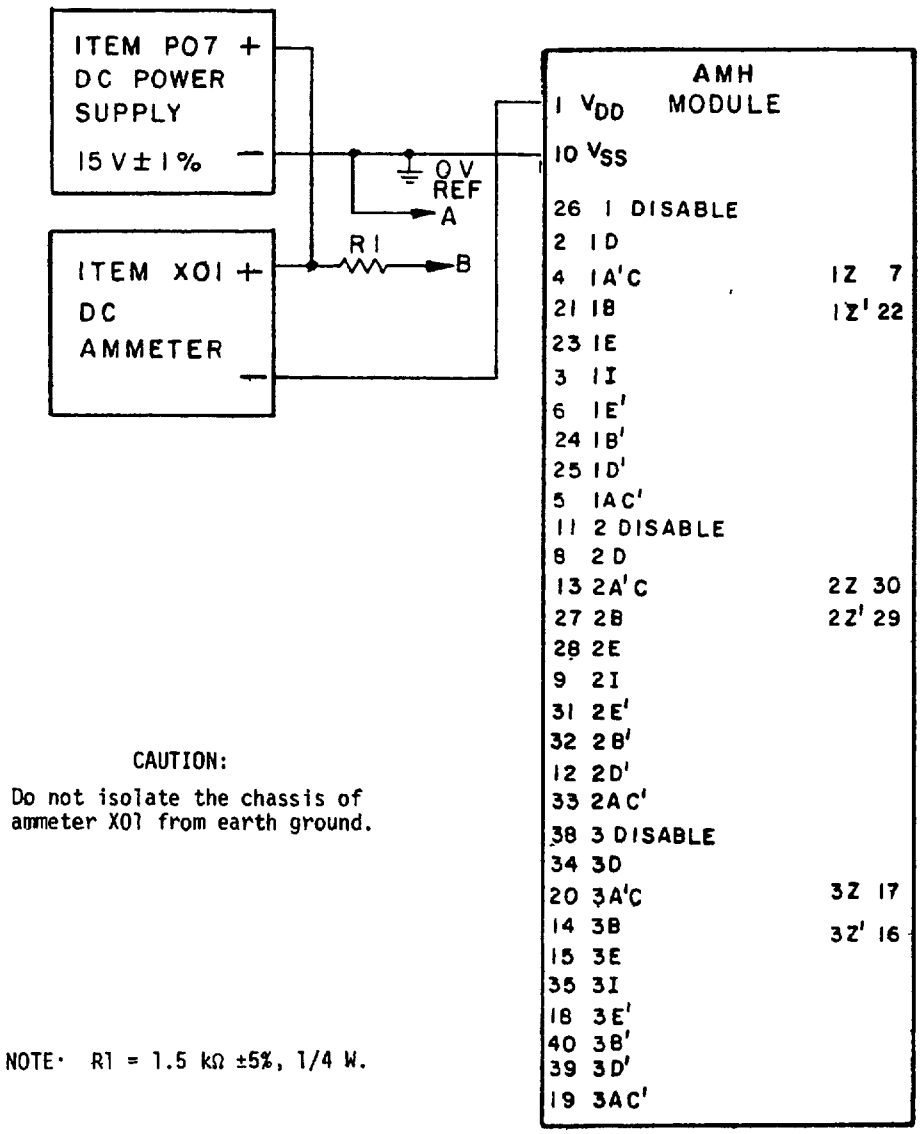
FIGURE 14. Input current (low-level) test circuit for table XVI.

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TABLE XVII. Static power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	Static power supply current, low level	26,2,4,21,23,6, 24,25,5,11,8,13, 27,28,31,32,12, 33,38,34,20,14, 15,18,40,39,19	3,9,35	0	12	µA	15
Q2	Static power supply current, high level	26,2,4,21,3,24, 25,5,11,8,13,27, 9,32,12,33,38, 34,20,14,35,40, 39,19	23,6,28,31,15, 18	0	12	µA	15

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 15. Static power supply current (low-level and high-level) test circuit for table XVII.

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TABLE XVIII. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Z	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	26,2,4,21, 24,25,5,11, 8,32,12,33, 38,34,20,14, 13,27,40,39, 19	23,6,28,31, 15,18	3,9,35	7	35	350	ns	16,17
R2	1Z'					22				
R3	2Z					30				
R4	2Z'					29				
R5	3Z					17				
R6	3Z'					16				
R7	1Z	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V				7	83	830		
R8	1Z'					22				
R9	2Z					30				
R10	2Z'					29				
R11	3Z					17				
R12	3Z'					16				

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TABLE XIX. Transition time (t_{THL}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1Z	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	26,2,4,21, 24,25,5,11, 8,32,12,33, 36,34,20,14, 13,27,40,39, 19	23,6,28,31, 15,18	3,9,35	7	17	175	ns	16,17
S2	1Z'					22				
S3	2Z					30				
S4	2Z'					29				
S5	3Z					17				
S6	3Z'					16				
S7	1Z	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V				7	65	655		
S8	1Z'					22				
S9	2Z					30				
S10	2Z'					29				
S11	3Z					17				
S12	3Z'					16				

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TABLE XX. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T1	1D to 1Z	26,21,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	44	440	ns	16,17
T2	1D' to 1Z'	26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
T3	2D to 2Z	26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
T4	2D' to 2Z'	26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
T5	3D to 3Z	26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
T6	3D' to 3Z'	26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
T7	1A'C to 1Z	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
T8	1A'C to 1Z'	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
T9	2A'C to 2Z	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
T10	2A'C to 2Z	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T11	3A'C to 3Z	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	44	440	ns	16,17
T12	3A'C to 3Z'	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
T13	1AC' to 1Z	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
T14	1AC' to 1Z'	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
T15	2AC' to 2Z	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
T16	2AC' to 2Z'	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
T17	3AC' to 3Z	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
T18	3AC' to 3Z'	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
T19	1B to 1Z	26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
T20	1B' to 1Z'	26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

Parameter: Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T21	2B to 2Z	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	44	440	ns	16,17
T22	2B' to 2Z'		26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
T23	3B to 3Z		26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
T24	3B' to 3Z'		26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
T25	1E to 1Z		26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	27	270		16,18
T26	1E' to 1Z'		26,2,4,21,23,3,24,25,5,11,8,13,27,26,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
T27	2E to 2Z		26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,36,34,20,14,15,35,18,40,39		28	30				
T28	2E' to 2Z'		26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T29	3E to 3Z	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	27	270	ns	16,18
T30	3E' to 3Z'	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	27	270		16,18
T31	1I to 1Z	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	36	360		16,17
T32	1I to 1Z'	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
T33	2I to 2Z	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
T34	2I to 2Z'	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
T35	3I to 3Z	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
T36	3I to 3Z'	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T37	1D to 1Z	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	26,21,3,24,25,11, 8,27,9,32,12, 38,34,14,35,40,39,	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	64	640	ns	16,17
T38	1D' to 1Z'		26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
T39	2D to 2Z		26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
T40	2D' to 2Z'		26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
T41	3D to 3Z		26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
T42	3D' to 3Z'		26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
T43	1A'C to 1Z		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
T44	1A'C to 1Z'		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
T45	2A'C to 2Z		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
T46	2A'C to 2Z'		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
T47	3A'C to 3Z	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	26,4,21,3,25,5,11, 13,27,9,12,33, 38,14,35,39,19	2,23,6,24,8,28,31, 32,34,15,16,40	20	17	64	640	ns	16,17
T48	3A'C to 3Z'		26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,16,40	20	16				
T49	1AC' to 1Z		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
T50	1AC' to 1Z'		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
T51	2AC' to 2Z		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
T52	2AC' to 2Z'		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
T53	3AC' to 3Z		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
T54	3AC' to 3Z'		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
T55	1B to 1Z		26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
T56	1B' to 1Z'		26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T57	2B to 2Z	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	64	640	ns	16,17
T58	2B' to 2Z'	26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
T59	3B to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
T60	3B' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
T61	1E to 1Z	26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	47	470		16,18
T62	1E' to 1Z'	26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
T63	2E to 2Z	26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
T64	2E' to 2Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
T65	3E to 3Z	26,2,4,21,23,3,6, 24,25,5,11,8, 13,27,28,9,31,32, 12,33,38,34,20,14, 35,18,40,39	Open	15	17	47	470	ns	16,18
T66	3E' to 3Z'	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	47	470		16,18
T67	1I to 1Z	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	56	560		16,17
T68	1I to 1Z'	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
T69	2I to 2Z	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
T70	2I to 2Z'	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,36,34,20,14,35, 40,39,19		9	29				
T71	3I to 3Z	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
T72	3I to 3Z'	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

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TABLE XXI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1D to 1Z	26,21,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	36	360	ns	16,17
U2	1D' to 1Z'	26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
U3	2D to 2Z	26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
U4	2D' to 2Z'	26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
U5	3D to 3Z	26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
U6	3D' to 3Z'	26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
U7	1A'C to 1Z	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
U8	1A'C to 1Z'	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
U9	2A'C to 2Z	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
U10	2A'C to 2Z'	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

Parameter: Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U11	3A'C to 3Z	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	36	360	ns	16,17
U12	3A'C to 3Z'	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
U13	1AC' to 1Z	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
U14	1AC' to 1Z'	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
U15	2AC' to 2Z	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
U16	2AC' to 2Z'	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
U17	3AC' to 3Z	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
U18	3AC' to 3Z'	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
U19	1B to 1Z	26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
U20	1B' to 1Z'	26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
U21	2B to 2Z	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	36	360	ns	16,17
U22	2B' to 2Z'	26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
U23	3B to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
U24	3B' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
U25	1E to 1Z	26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	27	270		16,18
U26	1E' to 1Z'	26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
U27	1E' to 1Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
U28	2E' to 2Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U29	3E to 3Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	27	270	ns	16,18
U30	3E' to 3Z'		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	27	270		16,18
U31	1I to 1Z		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	36	360		16,17
U32	1I to 1Z'		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
U33	2I to 2Z		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
U34	2I to 2Z'		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
U35	3I to 3Z		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
U36	3I to 3Z'		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
U37	1D to 1Z	26,21,3,24,25,11, 8,27,9,32,12, 38,34,14,35,40,39,	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	56	560	ns	16,17
U38	1D' to 1Z'	26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
U39	2D to 2Z	26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
U40	2D' to 2Z'	26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
U41	3D to 3Z	26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
U42	3D' to 3Z'	26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
U43	1A'C to 1Z	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
U44	1A'C to 1Z'	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
U45	2A'C to 2Z	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
U46	2A'C to 2Z'	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U47	3A'C to 3Z	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	26,4,21,3,25,5,11, 13,27,9,12,33, 38,14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	56	560	ns	16,17
U48	3A'C to 3Z'		26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
U49	1AC' to 1Z		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
U50	1AC' to 1Z'		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
U51	2AC' to 2Z		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
U52	2AC' to 2Z'		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
U53	3AC' to 3Z		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
U54	3AC' to 3Z'		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
U55	1B to 1Z		26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
U56	1B' to 1Z'		26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

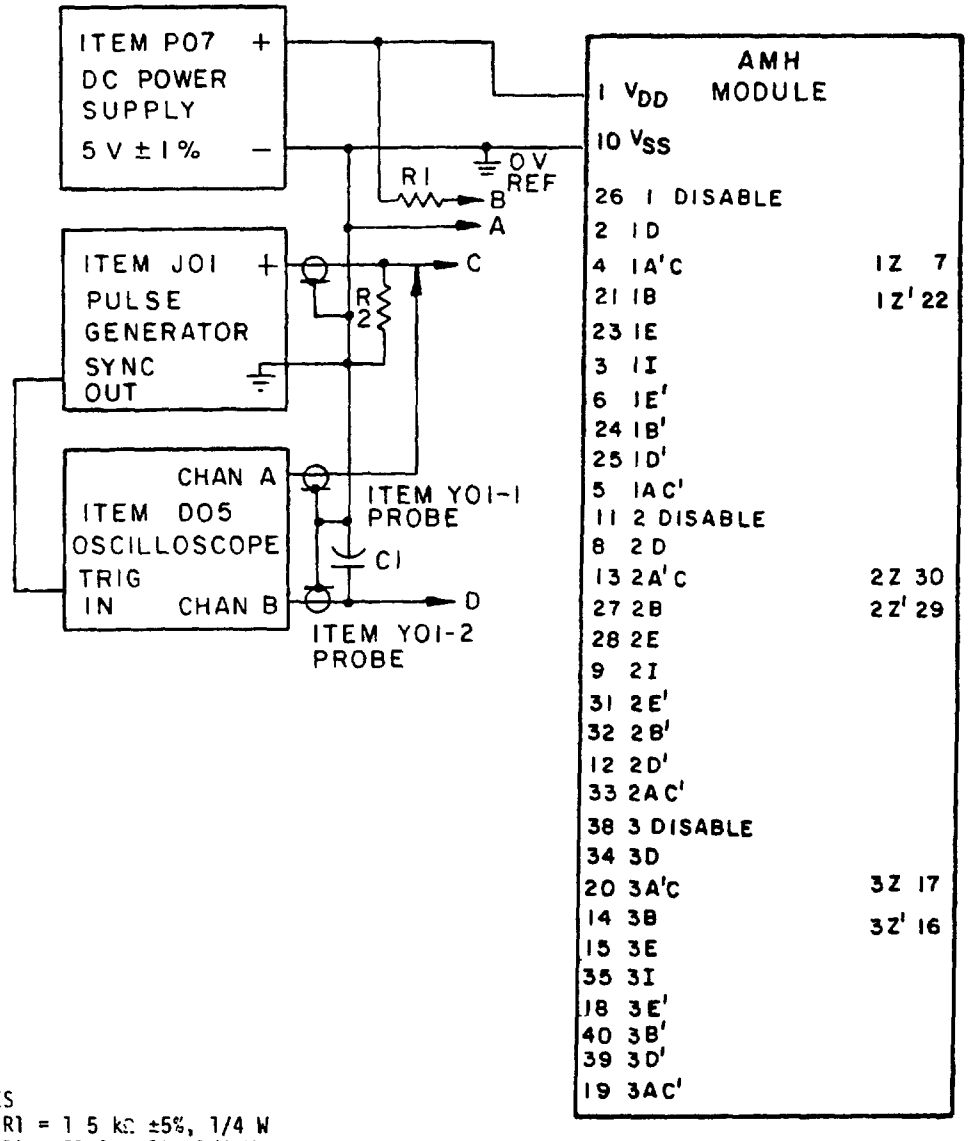
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U57	2B to 2Z	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	56	560	ns	16,17
U58	2B' to 2Z'	26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
U59	3B to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
U60	3B' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
U61	1E to 1Z	26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	47	470		16,18
U62	1E' to 1Z'	26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
U63	2E to 2Z	26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
U64	2E' to 2Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XXI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U65	3E to 3Z	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	26,2,4,21,23,3,6, 24,25,5,11,8, 13,27,28,9,31,32, 12,33,38,34,20,14, 35,18,40,39	Open	15	17	47	470	ns	16,18
U66	3E' to 3Z'		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	47	470		16,18
U67	1I to 1Z		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	56	560		16,17
U68	1I to 1Z'		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
U69	2I to 2Z		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
U70	2I to 2Z'		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
U71	3I to 3Z		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
U72	3I to 3Z'		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

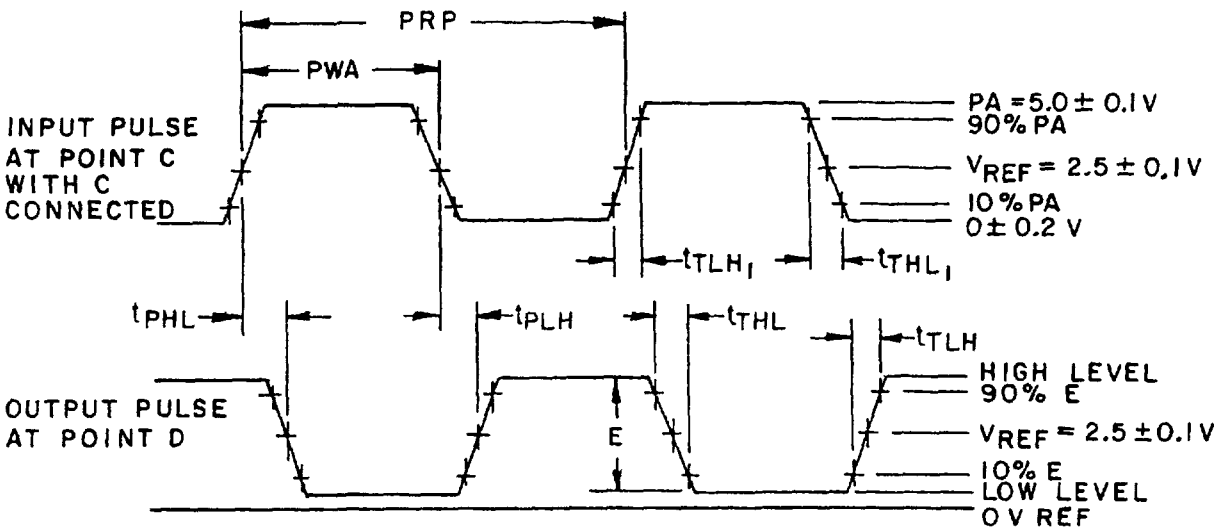
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NOTES
1. R1 = 1 5 kΩ ±5%, 1/4 W
2. R2 = 51.1Ω ±5%, 1/2 W
3. C1 = 50 ±5 pF (including probe and parasitic capacitance) for test conditions R1-R6, S1-S6, T1-T36, and U1-U36.
   = 150 ±15 pF (including probe and parasitic capacitance) for test conditions R7-R12, S7-S12, T37-T72, and U37-U72.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVIII through XXI.

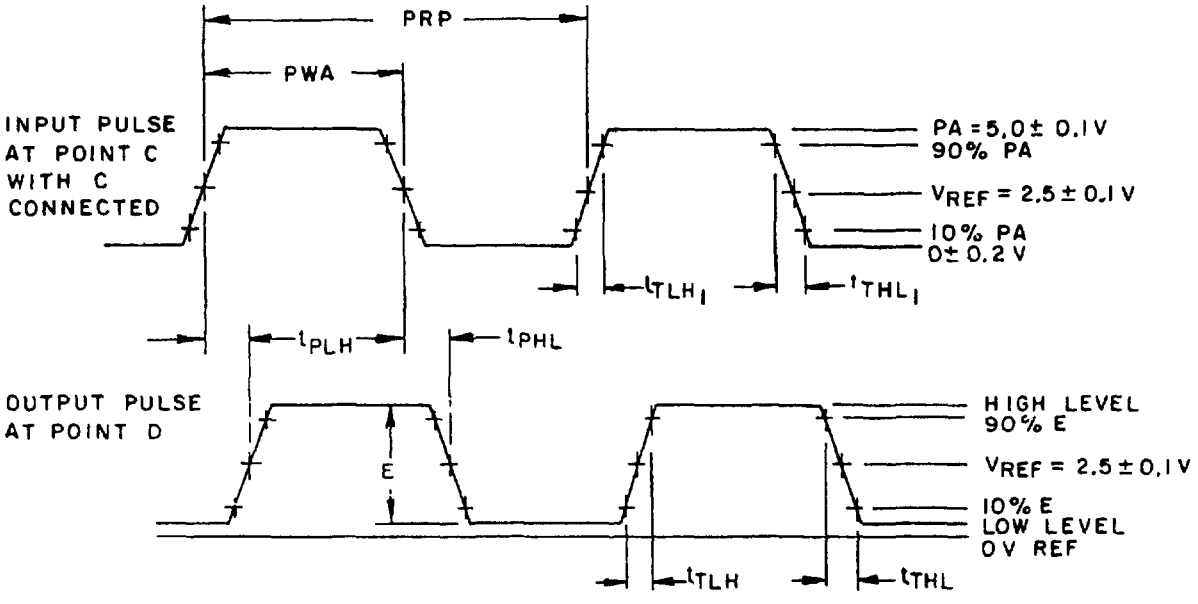
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:04Z
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0 5 µs.
2. Before making measurements verify that the output pulse high level is ≥ 4 6 V and the low level is ≤ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm)
4. All voltages are referenced to V_{SS}.

FIGURE 17 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XVIII through XXI.

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NOTES
1. Input pulse parameters
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns
   PWA = 2.5 ±0.2 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making measurements verify that the output pulse high level is $\geq$ 4.6 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}

FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XX and XXI.

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TABLE XXII. Disable delay time (t_{PZH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 5 V	Open	23,6,28,31,15,18	26	7	25	250	ns	19, 20
V2				26	22				
V3				11	30				
V4				11	29				
V5				38	17				
V6				38	16				
V7	Disable delay time (t_{PZH}), C_L = 150 pF, V_{DD} = 5 V			26	7	45	450		
V8				26	22				
V9				11	30				
V10				11	29				
V11				38	17				
V12				38	16				

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TABLE XXIII. Disable delay time (t_{PHZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 5 V	Open	23,6,28,31,15,18	26	7	25	250	ns	19,20
W2				26	22				
W3				11	30				
W4				11	29				
W5				38	17				
W6				38	16				
W7	Disable delay time (t_{PHZ}), C_L = 150 pF, V_{DD} = 5 V			26	7	45	450		
W8				26	22				
W9				11	30				
W10				11	29				
W11				38	17				
W12				38	16				

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TABLE XXIV. Disable delay time (t_{PZL}) test.

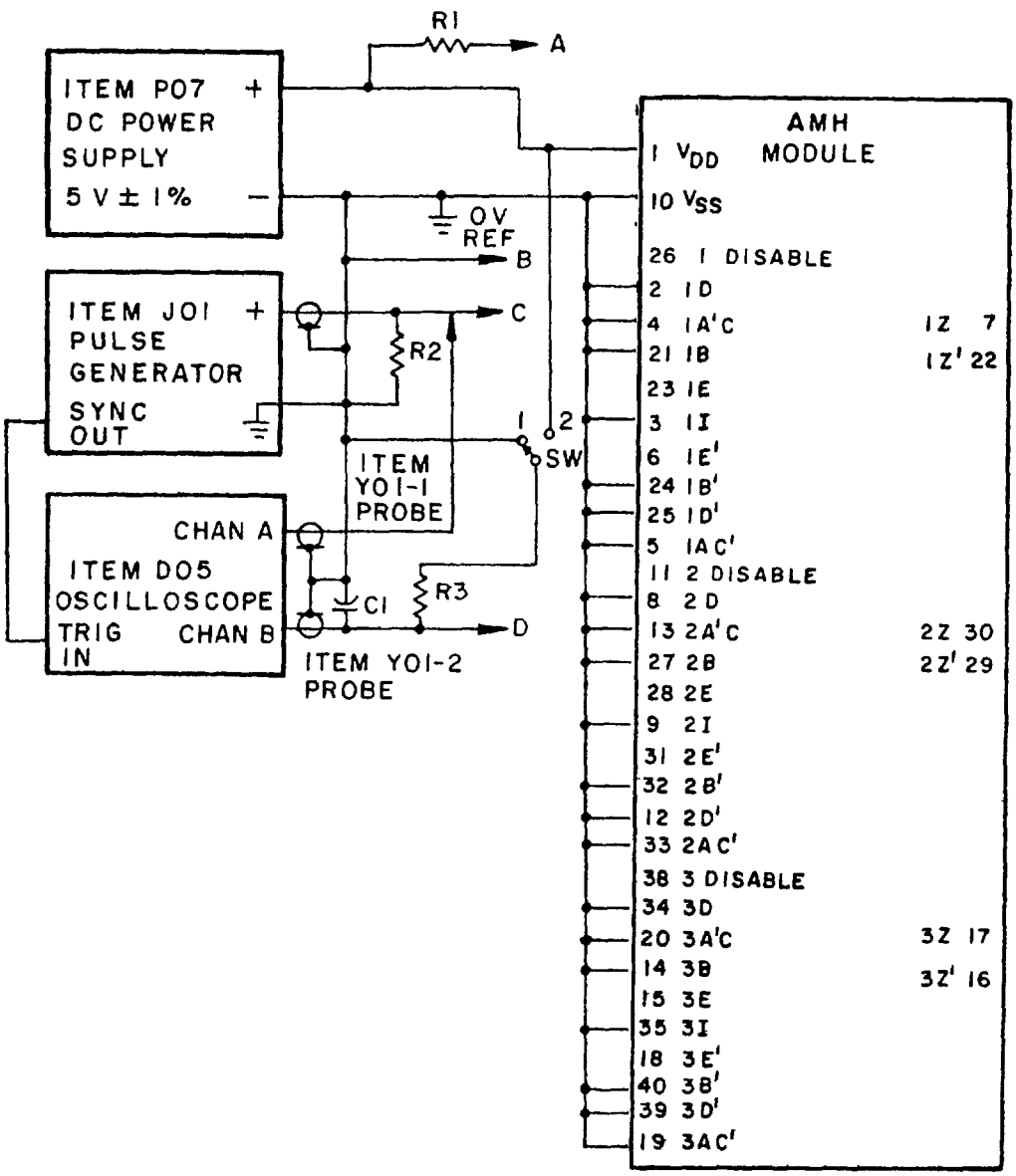
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X1	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 5 V	23,6,28,31,15,18	Open	26	7	13	135	ns	19,20
X2				26	22				
X3				11	30				
X4				11	29				
X5				38	17				
X6				38	16				
X7	Disable delay time (t_{PZL}), C_L = 150 pF, V_{DD} = 5 V			26	7	33	335		
X8				26	22				
X9				11	30				
X10				11	29				
X11				38	17				
X12				38	16				

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TABLE XXV. Disable delay time (t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 5 V	23,6,28,31,15,18	Open	26	7	25	255	ns	19,20
Y2				26	22				
Y3				11	30				
Y4				11	29				
Y5				38	17				
Y6				38	16				
Y7	Disable delay time (t_{PLZ}), C_L = 150 pF, V_{DD} = 5 V			26	7	45	455		
Y8				26	22				
Y9				11	30				
Y10				11	29				
Y11				38	17				
Y12				38	16				

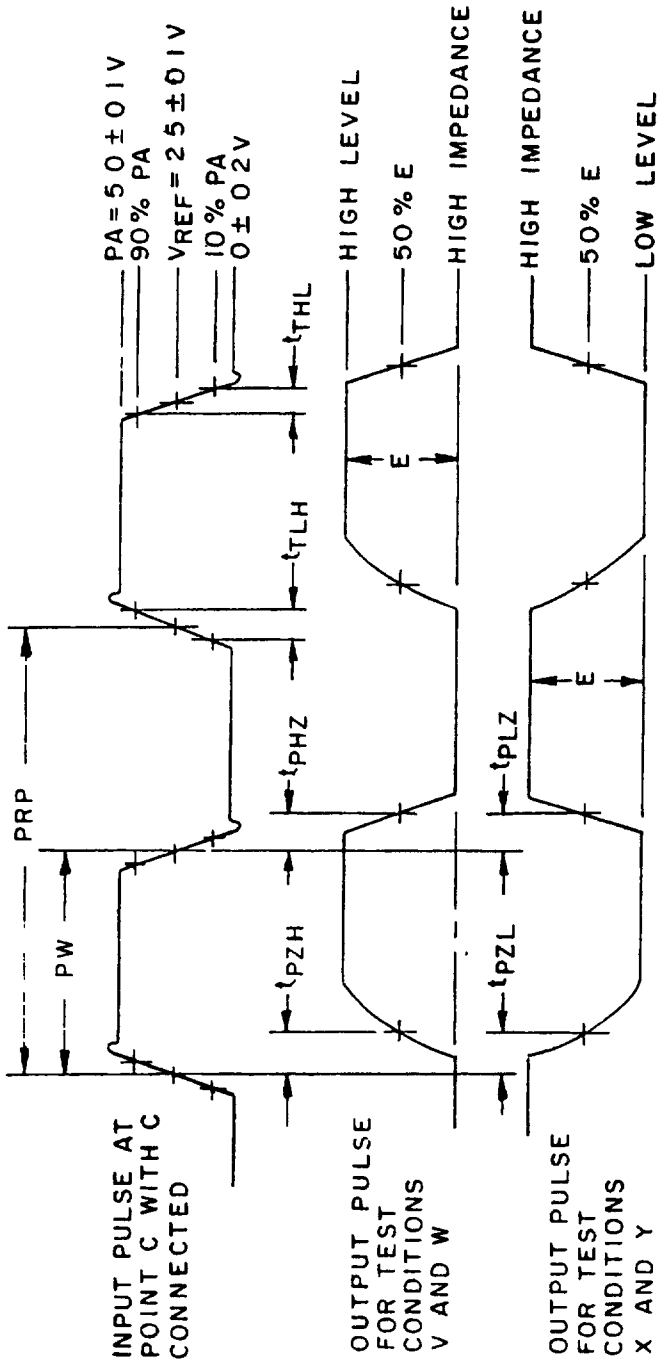
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.
3. R3 = 1.0 kΩ ±1%, 1/4 W.
4. C1 = 50 pF ±10% for test conditions V1-V6, W1-W6, X1-X6, and Y1-Y6.
   = 150 pF ±10% for test conditions V7-V12, W7-W12, X7-X12, and Y7-Y12
5 Place switch SW1 in position 1 for test conditions X1-X12 and Y1-Y12, and in position 2 for test conditions V1-V12 and W1-W12.

FIGURE 19. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit for tables XXII through XXV.

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NOTES
1. Input pulse parameters:
   PW = 8 ±0.5 µs
   PRP = 16 ±1.0 µs.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns
2. Before making the measurements specified verify the output pulse high level is ≥ 4.6 V and the low level is ≤ 0.4 V.

FIGURE 20. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit waveforms for tables XXII through XXV.

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TABLE XXVI. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Z1	1Z	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	26,2,4,21,24,25,5, 11,8,32,12,33,38,34, 20,14,13,27,40,39,19	23,6,28,31,15,18	3,9,35	7	15	150	ns	21,22
Z2	1Z'					22				
Z3	2Z					30				
Z4	2Z'					29				
Z5	3Z					17				
Z6	3Z'					16				

TABLE XXVII. Transition time (t_{THL}) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
AA1	1Z	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	26,2,4,21,24,25,5, 11,8,32,12,33,38,34, 20,14,13,27,40,39,19	23,6,28,31,15,18	3,9,35	7	7	75	ns	21,22
AA2	1Z'					22				
AA3	2Z					30				
AA4	2Z'					29				
AA5	3Z					17				
AA6	3Z'					16				

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TABLE XXVIII. Propagation delay time (t_{PLH}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB1	1D to 1Z	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	26,21,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	16	165	ns	21,22
BB2	1D' to 1Z'		26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
BB3	2D to 2Z		26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
BB4	2D' to 2Z'		26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
BB5	3D to 3Z		26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
BB6	3D' to 3Z'		26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
BB7	1A'C to 1Z		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
BB8	1A'C to 1Z'		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
BB9	2A'C to 2Z		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
BB10	2A'C to 2Z'		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

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TABLE XXVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
BB11	3A'C to 3Z	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	16	165	ns	21,22
BB12	3A'C to 3Z'	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
BB13	1AC' to 1Z	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
BB14	1AC' to 1Z'	26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
BB15	2AC' to 2Z	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
BB16	2AC' to 2Z'	26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
BB17	3AC' to 3Z	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
BB18	3AC' to 3Z'	26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
BB19	1B to 1Z	26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
BB20	1B' to 1Z'	26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V

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TABLE XXVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB21	2B to 2Z	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,19	4,23,6,5,13,28,31,33,20,15,18,19	27	30	16	165	ns	21,22
BB22	2B' to 2Z'		26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
BB23	3B to 3Z		26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
BB24	3B' to 3Z'		26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
BB25	1E to 1Z		26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	11	110		21,23
BB26	1E' to 1Z'		26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
BB27	2E to 2Z		26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
BB28	2E' to 2Z'		26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XXVIII. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB29	3E to 3Z	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	11	110	ns	21,23
BB30	3E' to 3Z'		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	11	110		21,23
BB31	1I to 1Z		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	15	150		21,22
BB32	1I to 1Z'		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
BB33	2I to 2Z		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
BB34	2I to 2Z'		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
BB35	3I to 3Z		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
BB36	3I to 3Z'		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

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TABLE XXIX. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
CC1	1D to 1Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	26,21,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	13	135	ns	21,22
CC2	1D' to 1Z'		26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
CC3	2D to 2Z		26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
CC4	2D' to 2Z'		26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
CC5	3D to 3Z		26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
CC6	3D' to 3Z'		26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,40		39	16				
CC7	1A'C to 1Z		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
CC8	1A'C to 1Z'		26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
CC9	2A'C to 2Z		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
CC10	2A'C to 2Z'		26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

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TABLE XXIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
CC11	3A'C to 3Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	13	135	ns	21,22
CC12	3A'C to 3Z'		26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
CC13	1AC' to 1Z		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
CC14	1AC' to 1Z'		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
CC15	2AC' to 2Z		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
CC16	2AC' to 2Z'		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
CC17	3AC' to 3Z		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
CC18	3AC' to 3Z'		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
CC19	1B to 1Z		26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
CC20	1B' to 1Z'		26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

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TABLE XXIX. Propagation delay time (t_{PHL}) test - Continued.

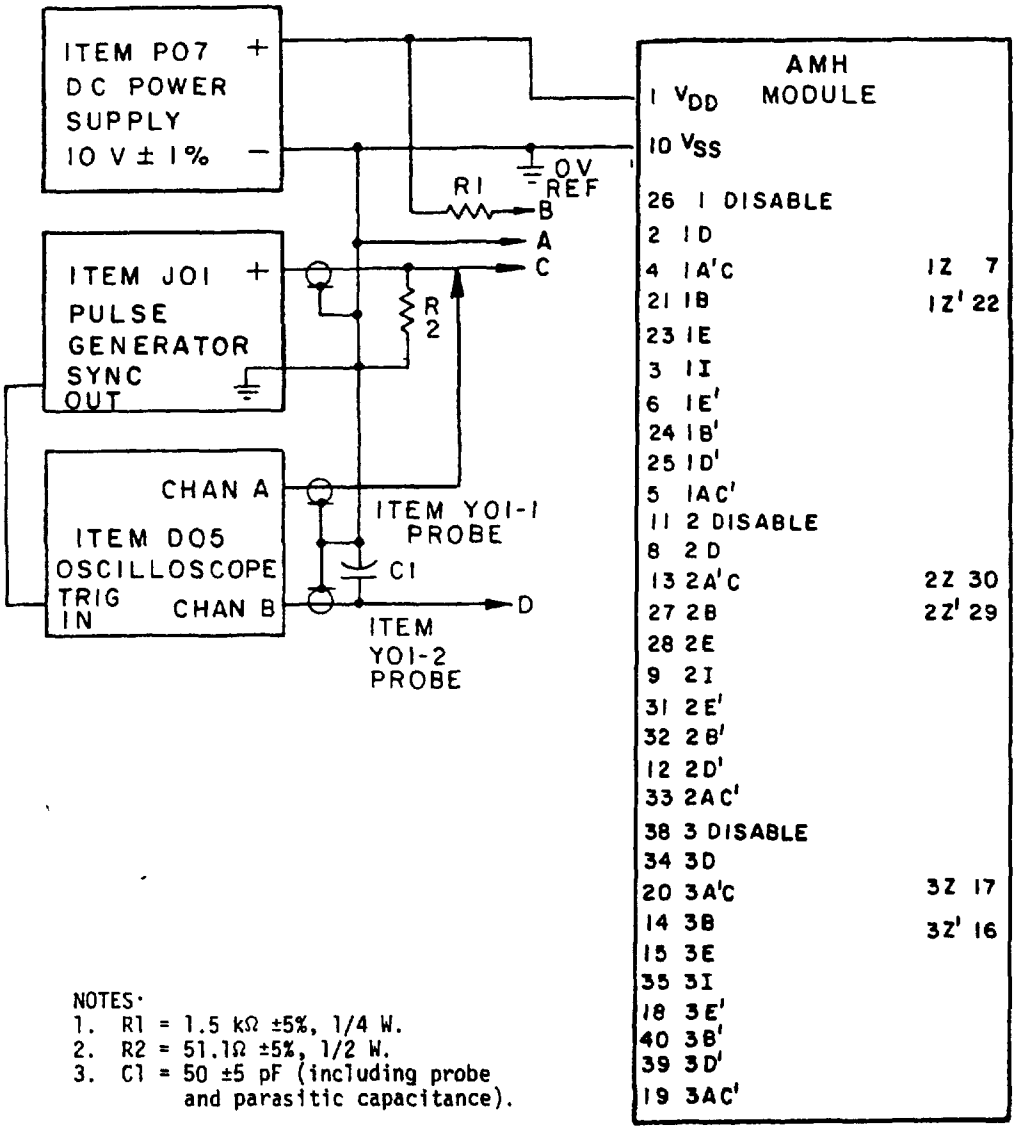
Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
CC21	2B to 2Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	26,2,21,3,24,25,11, 8,9,32,12,38,34,14, 35,40,19	4,23,6,5,13,28,31, 33,20,15,18,19	27	30	13	135	ns	21,22
CC22	2B' to 2Z'		26,2,21,3,24,25,11, 8,27,9,12,38,34,14, 35,40,39		32	29				
CC23	3B to 3Z		26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 35,40,39		14	17				
CC24	3B' to 3Z'		26,2,21,3,24,25,11, 8,27,9,32,12,38,34, 14,35,39		40	16				
CC25	1E to 1Z		26,2,4,21,3,6,24, 25,5,11,8,13,27,28, 9,19,31,32,12,33, 38,34,20,14,15,35, 18,40,39	Open	23	7	9	90		21,23
CC26	1E' to 1Z'		26,2,4,21,23,3,24, 25,5,11,8,13,27,28, 9,19,31,32,12,33, 38,34,20,14,15,35, 18,40,39		6	22				
CC27	2E to 2Z		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 9,19,31,32,12,33, 38,34,20,14,15,35, 18,40,39		28	30				
CC28	2E' to 2Z'		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,32,12,33, 38,34,20,14,15,35, 18,40,39		31	29				

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TABLE XXIX. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect		C to pin	D to pin	Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin			Min	Max		
CC29	3E to 3Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	9	90	ns	21,23
CC30	3E' to 3Z'		26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	9	90		21,23
CC31	1I to 1Z		26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	14	140		21,22
CC32	1I to 1Z'		26,2,24,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
CC33	2I to 2Z		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
CC34	2I to 2Z'		26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
CC35	3I to 3Z		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
CC36	3I to 3Z'		26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

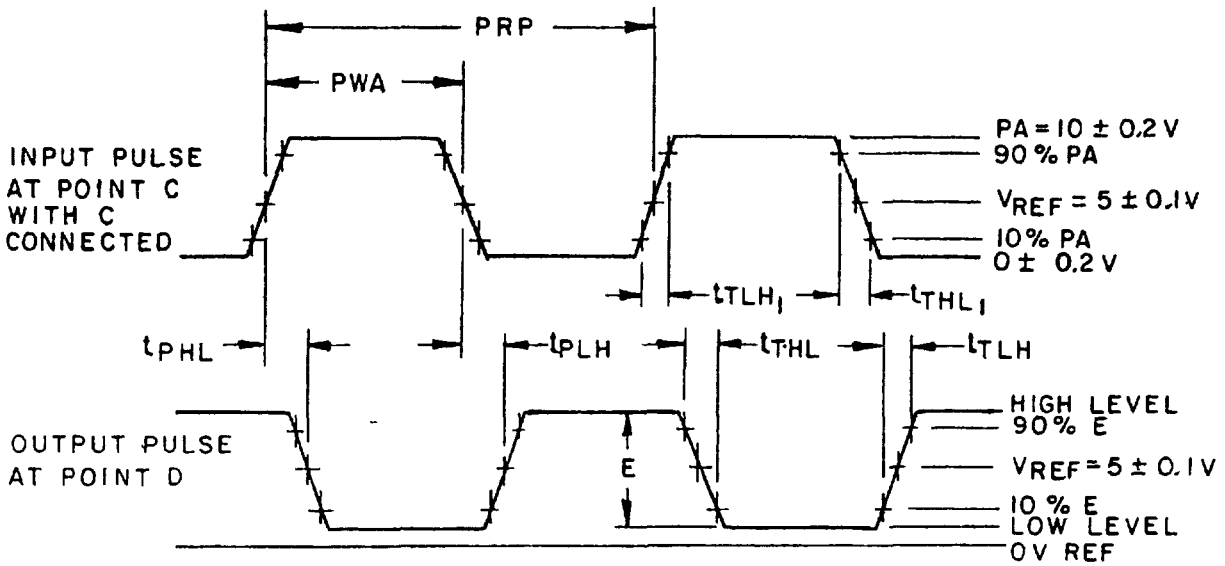
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NOTES:
1. R1 = 1.5 kΩ ±5%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.
3. C1 = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 21. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXVII through XXIX.

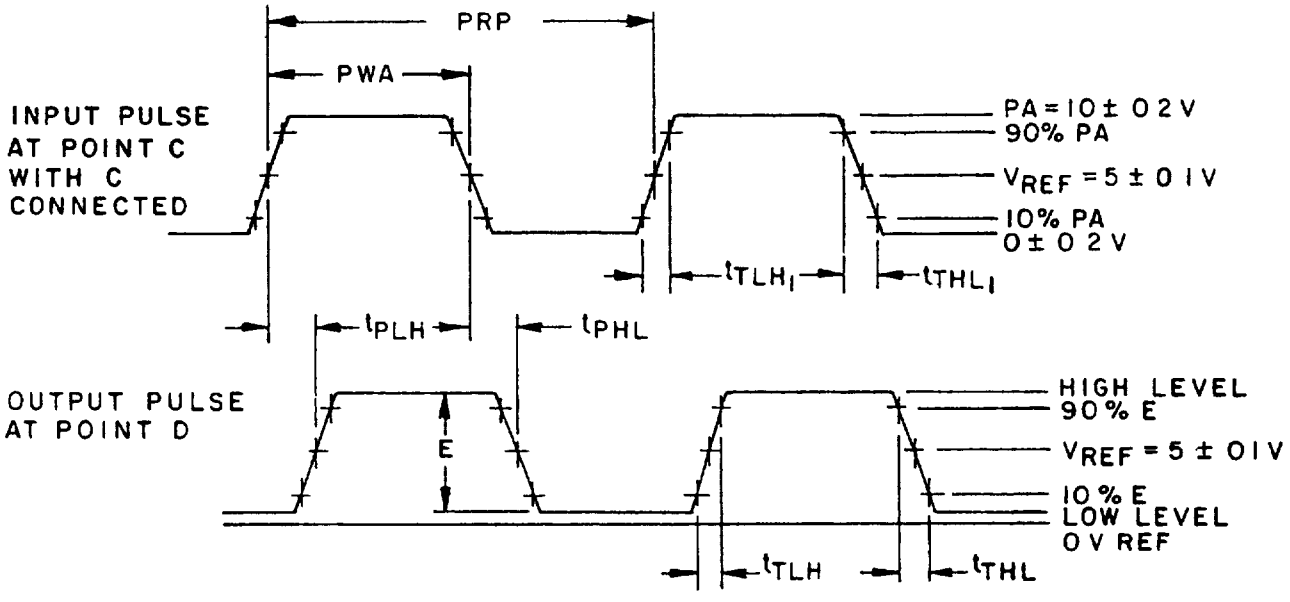
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NOTES
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs
   PRP = 5.0 ±0.5 µs
2. Before making measurements specified in tables XIX through XXII, verify that the output pulse high level is ≥ 9.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1 and YO1-2 to input and output pins shall be less than 1.00 inch (25.4 mm)
4. All voltages are referenced to V_{SS}.

FIGURE 22. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXVII through XXIX.

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NOTES
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs
   PRP = 5.0 ±0.5 µs.
2. Before making measurements specified in tables XXI through XXIV, verify that the output pulse high level is $\geq$ 9.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes Y01-1 and Y01-2 to input and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 23 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXVII through XXIX.

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TABLE XXX. Disable delay time (t_{PZH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
DD1	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 10 V	Open	23,6,28,31,15,18	26	7	10	100	ns	24,25
DD2	↓	↓	↓	26	22	↓	↓	↓	↓
DD3				11	30				
DD4				11	29				
DD5				38	17				
DD6				38	16				

TABLE XXXI. Disable delay time (t_{PHZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
EE1	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 10 V	Open	23,6,28,31,15,18	26	7	10	100	ns	24,25
EE2	↓	↓	↓	26	22	↓	↓	↓	↓
EE3				11	30				
EE4				11	29				
EE5				38	17				
EE6				38	16				

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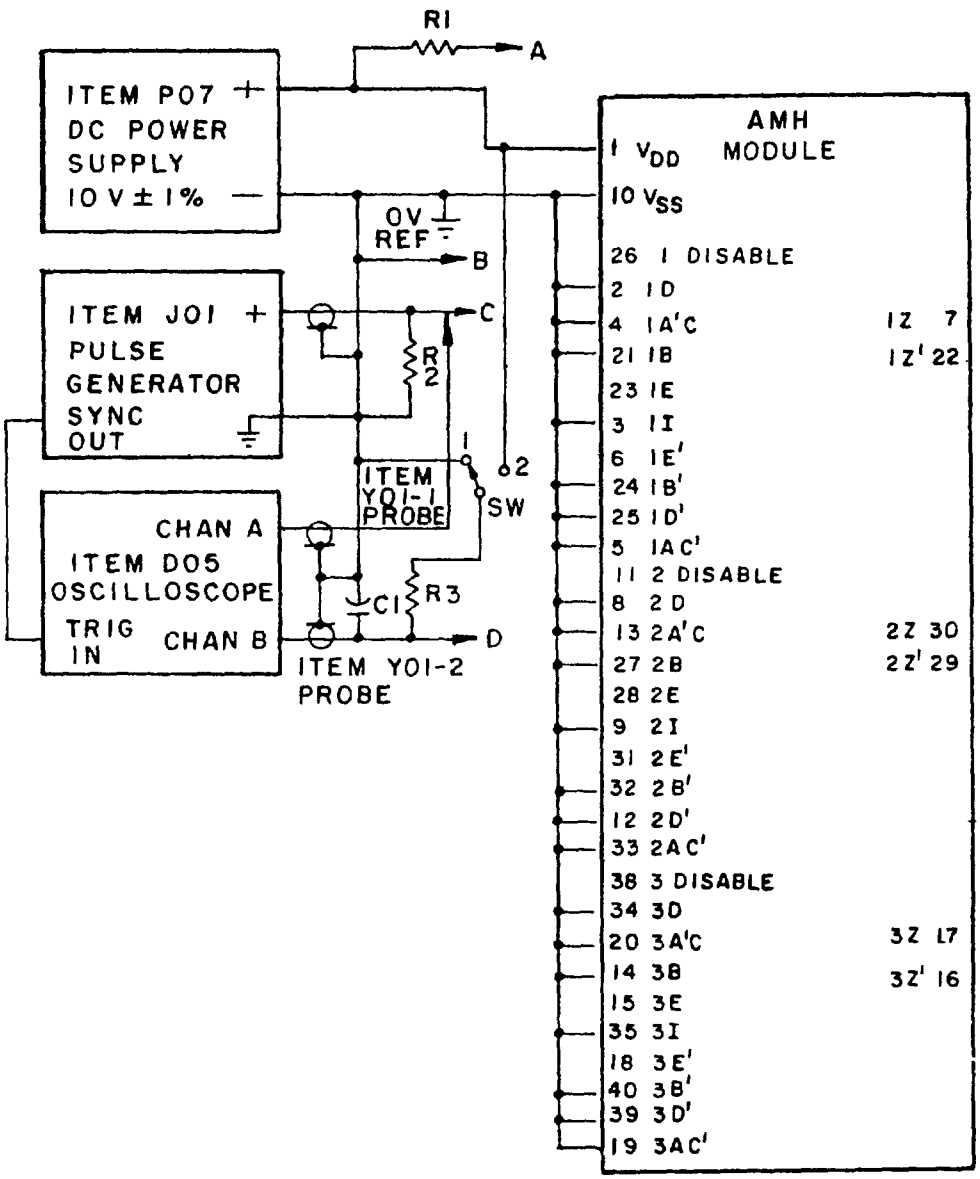
TABLE XXXII. Disable delay time (t_{PZL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 10 V	23,6,28,31,15,18	Open	26	7	10	100	ns	24,25
FF2				26	22				
FF3				11	30				
FF4				11	29				
FF5				38	17				
FF6				38	16				

TABLE XXXIII. Disable delay time (t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
GG1	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 10 V	23,6,28,31,15,18	Open	26	7	10	105	ns	24,25
GG2				26	22				
GG3				11	30				
GG4				11	29				
GG5				38	17				
GG6				38	16				

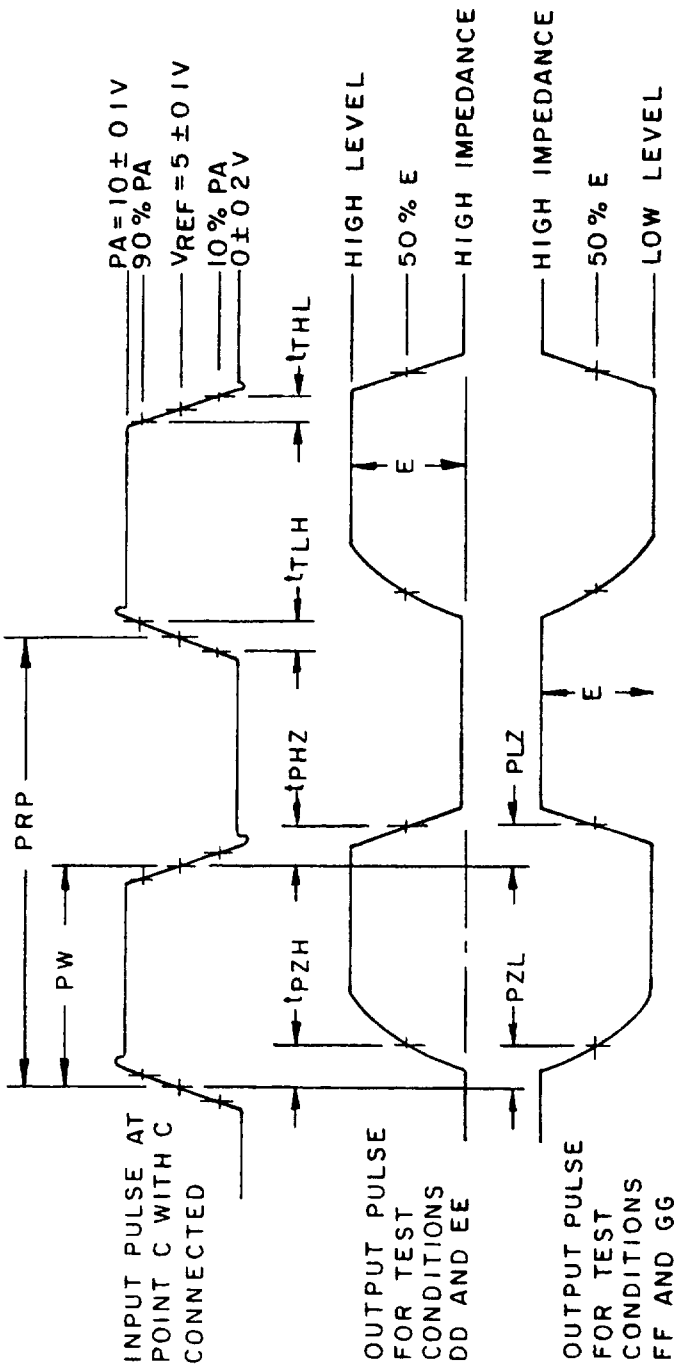
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NOTES.
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.
3. R3 = 1.0 kΩ ±1%, 1/4 W.
4. C1 = 50 pF ±10% (including probe and parasitic capacitance).
5. Place switch SW1 in position 1 for test conditions FF1-FF6 and GG1-GG6, and in position 2 for test conditions DD1-DD6 and EE1-EE6.

FIGURE 24. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit for tables XXX through XXXIII.

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NOTES
1. Input pulse parameters
   PW = 8.0 ± 0.5 µs
   PRP = 16.1.0 µs
   $t_{TLH} = t_{THL}$ = 20 ± 2 ns
2. Before making the measurements specified verify the output pulse high level is ≥ 9.5 V and the low level is ≤ 0.5 V.

FIGURE 25. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit waveforms for tables XXX through XXXIII.

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TABLE XXXIV. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
HH1	1Z Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	26,2,4,21, 24,25,5,11, 8,32,12,33, 38,34,20,14, 13,27,40,39, 19	23,6,28,31, 15,18	3,9,35	7	11	110	ns	26,27
HH2	1Z'				22				
HH3	2Z				30				
HH4	2Z'				29				
HH5	3Z				17				
HH6	3Z'				16				

TABLE XXXV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
JJ1	1Z Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	26,2,4,21, 24,25,5,11, 8,32,12,33, 38,34,20,14, 13,27,40,39, 19	23,6,28,31, 15,18	3,9,35	7	5	55	ns	26,27
JJ2	1Z'				22				
JJ3	2Z				30				
JJ4	2Z'				29				
JJ5	3Z				17				
JJ6	3Z'				16				

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TABLE XXXVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°+5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
KK1	1D to 1Z	26,21,3,24,25,11,8,27,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	2	7	11	110	ns	26,27
KK2	1D' to 1Z'	26,2,21,3,24,11,8,27,9,32,12,38,34,14,35,40,39		25	22				
KK3	2D to 2Z	26,2,21,3,24,25,11,27,9,32,12,38,34,14,35,40,39		8	30				
KK4	2D' to 2Z'	26,2,21,3,24,25,11,8,27,9,32,38,34,14,35,40,39		12	29				
KK5	3D to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,14,35,40,39		34	17				
KK6	3D' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,40		39	16				
KK7	1A'C to 1Z	26,21,3,25,5,11,13,27,9,12,33,38,20,14,35,39,19	2,23,6,24,8,28,31,32,34,15,18,40	4	7				
KK8	1A'C to 1Z'	26,21,3,25,5,11,13,27,9,12,33,38,20,14,35,39,19		4	22				
KK9	2A'C to 2Z	26,4,21,3,25,5,11,27,9,12,33,38,20,14,35,39,19		13	30				
KK10	2A'C to 2Z'	26,4,21,3,25,5,11,27,9,12,33,38,20,14,35,39,19		13	29				

Parameter: Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V

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TABLE XXXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
KK11	3A'C to 3Z	26,4,21,3,25,5,11,13,27,9,12,33,38,14,35,39,19	2,23,6,24,8,28,31,32,34,15,18,40	20	17	11	110	ns	26,27
KK12	3A'C to 3Z'	26,4,21,3,25,5,11,13,27,9,12,33,38,14,35,39,19	2,23,6,24,8,28,31,32,34,15,18,40	20	16				
KK13	1AC' to 1Z	26,2,4,3,24,11,8,13,9,32,33,38,34,20,35,40,19	21,23,6,25,27,28,31,12,14,15,18,39	5	7				
KK14	1AC' to 1Z'	26,2,4,3,24,11,8,13,9,32,33,38,34,20,35,40,19		5	22				
KK15	2AC' to 2Z	26,2,4,3,24,5,11,8,13,9,32,38,34,20,35,40,19		33	30				
KK16	2AC' to 2Z'	26,2,4,3,24,5,11,8,13,9,32,38,34,20,35,40,19		33	29				
KK17	3AC' to 3Z	26,2,4,3,24,5,11,8,13,9,32,33,38,34,20,35,40		19	17				
KK18	3AC' to 3Z'	26,2,4,3,24,5,11,8,13,9,32,33,38,34,20,35,40		19	16				
KK19	1B to 1Z	26,2,3,24,25,11,8,27,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	21	7				
KK20	1B' to 1Z'	26,2,21,3,25,11,8,27,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	24	22				

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TABLE XXXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
KK21	2B to 2Z	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	11	110	ns	26,27
KK22	2B' to 2Z'	26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
KK23	3B to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
KK24	3B' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
KK25	1E to 1Z	26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	7	75		26,28
KK26	1E' to 1Z'	26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
KK27	2E to 2Z	26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
KK28	2E' to 2Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XXXVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
KK29	3E to 3Z	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	7	75	ns	26,28
KK30	3E' to 3Z'	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	7	75		26,28
KK31	1I to 1Z	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	10	100		26,27
KK32	1I to 1Z'	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
KK33	2I to 2Z	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
KK34	2I to 2Z'	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
KK35	3I to 3Z	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
KK36	3I to 3Z'	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

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TABLE XXXVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
LL1	1D to 1Z	26,21,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	2	7	11	110	ns	26,27
LL2	1D' to 1Z'	26,2,21,3,24,11,8, 27,9,32,12,38,34, 14,35,40,39		25	22				
LL3	2D to 2Z	26,2,21,3,24,25,11, 27,9,32,12,38,34, 14,35,40,39		8	30				
LL4	2D' to 2Z'	26,2,21,3,24,25,11, 8,27,9,32,38,34,14, 35,40,39		12	29				
LL5	3D to 3Z	26,2,21,3,24,25,11, 8,27,9,32,12,38,14, 35,40,39		34	17				
LL6	3D' to 3Z'	26,2,21,3,24,25,11, 6,27,9,32,12,38,34, 14,35,40		39	16				
LL7	1A'C to 1Z	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	4	7				
LL8	1A'C to 1Z'	26,21,3,25,5,11,13, 27,9,12,33,38,20, 14,35,39,19		4	22				
LL9	2A'C to 2Z	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	30				
LL10	2A'C to 2Z'	26,4,21,3,25,5,11, 27,9,12,33,38,20, 14,35,39,19		13	29				

Conditions: t_{PHL}, C_L = 50 pF, V_{DD} = 15 V

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TABLE XXXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
LL11	3A'C to 3Z	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	17	11	110	ns	26,27
LL12	3A'C to 3Z'		26,4,21,3,25,5,11, 13,27,9,12,33,38, 14,35,39,19	2,23,6,24,8,28,31, 32,34,15,18,40	20	16				
LL13	1AC' to 1Z		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19	21,23,6,25,27,28, 31,12,14,15,18,39	5	7				
LL14	1AC' to 1Z'		26,2,4,3,24,11,8, 13,9,32,33,38,34, 20,35,40,19		5	22				
LL15	2AC' to 2Z		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	30				
LL16	2AC' to 2Z'		26,2,4,3,24,5,11,8, 13,9,32,38,34,20, 35,40,19		33	29				
LL17	3AC' to 3Z		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	17				
LL18	3AC' to 3Z'		26,2,4,3,24,5,11,8, 13,9,32,33,38,34, 20,35,40		19	16				
LL19	1B to 1Z		26,2,3,24,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	21	7				
LL20	1B' to 1Z'		26,2,21,3,25,11,8, 27,9,32,12,38,34, 14,35,40,39	4,23,6,5,13,28,31, 33,20,15,18,19	24	22				

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TABLE XXXVII. Propagation delay time (t_{PHL}) test - Continued.

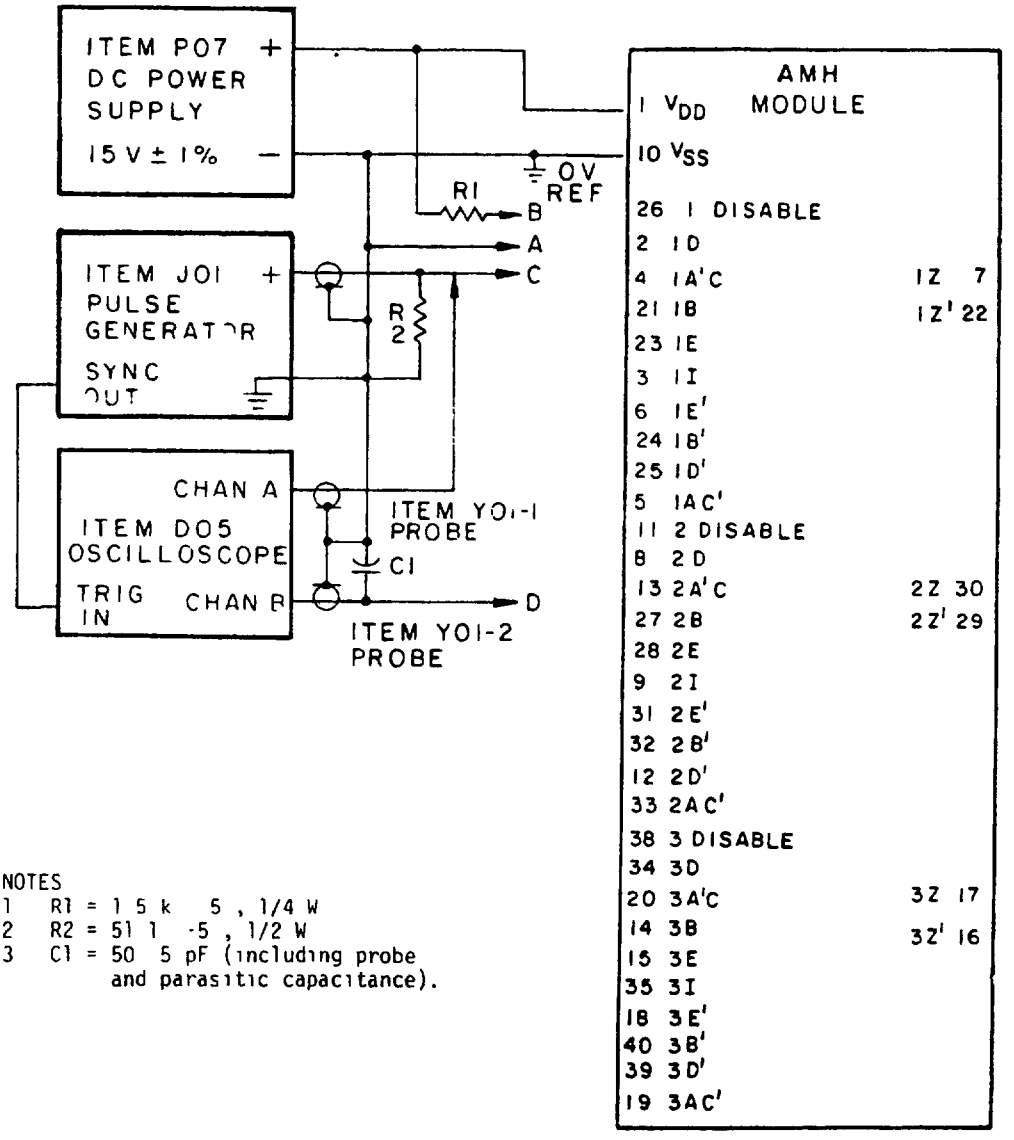
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
LL21	2B to 2Z	26,2,21,3,24,25,11,8,9,32,12,38,34,14,35,40,39	4,23,6,5,13,28,31,33,20,15,18,19	27	30	11	110	ns	26,27
LL22	2B' to 2Z'	26,2,21,3,24,25,11,8,27,9,12,38,34,14,35,40,39		32	29				
LL23	3B to 3Z	26,2,21,3,24,25,11,8,27,9,32,12,38,34,35,40,39		14	17				
LL24	3B' to 3Z'	26,2,21,3,24,25,11,8,27,9,32,12,38,34,14,35,39		40	16				
LL25	1E to 1Z	26,2,4,21,3,6,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39	Open	23	7	6	60		26,28
LL26	1E' to 1Z'	26,2,4,21,23,3,24,25,5,11,8,13,27,28,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		6	22				
LL27	2E to 2Z	26,2,4,21,23,3,6,24,25,5,11,8,13,27,9,19,31,32,12,33,38,34,20,14,15,35,18,40,39		28	30				
LL28	2E' to 2Z'	26,2,4,21,23,3,6,24,25,5,11,8,13,27,28,9,19,32,12,33,38,34,20,14,15,35,18,40,39		31	29				

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TABLE XXXVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
LL29	3E to 3Z	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,31,32,12,33, 38,34,20,14,35,18, 40,39	Open	15	17	6	60	ns	26,28
LL30	3E' to 3Z'	26,2,4,21,23,3,6, 24,25,5,11,8,13,27, 28,9,19,31,32,12, 33,38,34,20,14,15, 35,40,39	Open	18	16	6	60		26,28
LL31	1I to 1Z	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19	23,6,28,31,15,18	3	7	10	100		26,27
LL32	1I to 1Z'	26,2,4,21,24,25,5, 11,8,13,27,9,32,12, 33,38,34,20,14,35, 40,39,19		3	22				
LL33	2I to 2Z	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	30				
LL34	2I to 2Z'	26,2,4,21,3,24,25, 5,11,8,13,27,32,12, 33,38,34,20,14,35, 40,39,19		9	29				
LL35	3I to 3Z	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	17				
LL36	3I to 3Z'	26,2,4,21,3,24,25, 5,11,8,13,27,9,32, 12,33,38,34,20,14, 40,39,19		35	16				

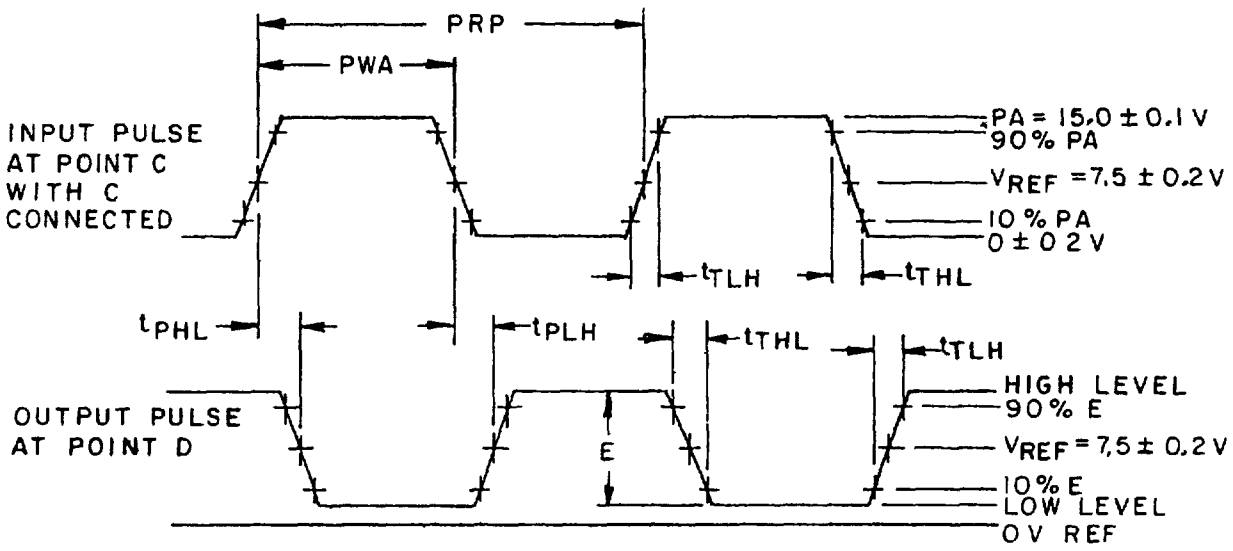
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NOTES
1 R1 = 1 5 k 5 , 1/4 W
2 R2 = 51 1 -5 , 1/2 W
3 C1 = 50 5 pF (including probe and parasitic capacitance).

FIGURE 26. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XXXIV through XXXVII

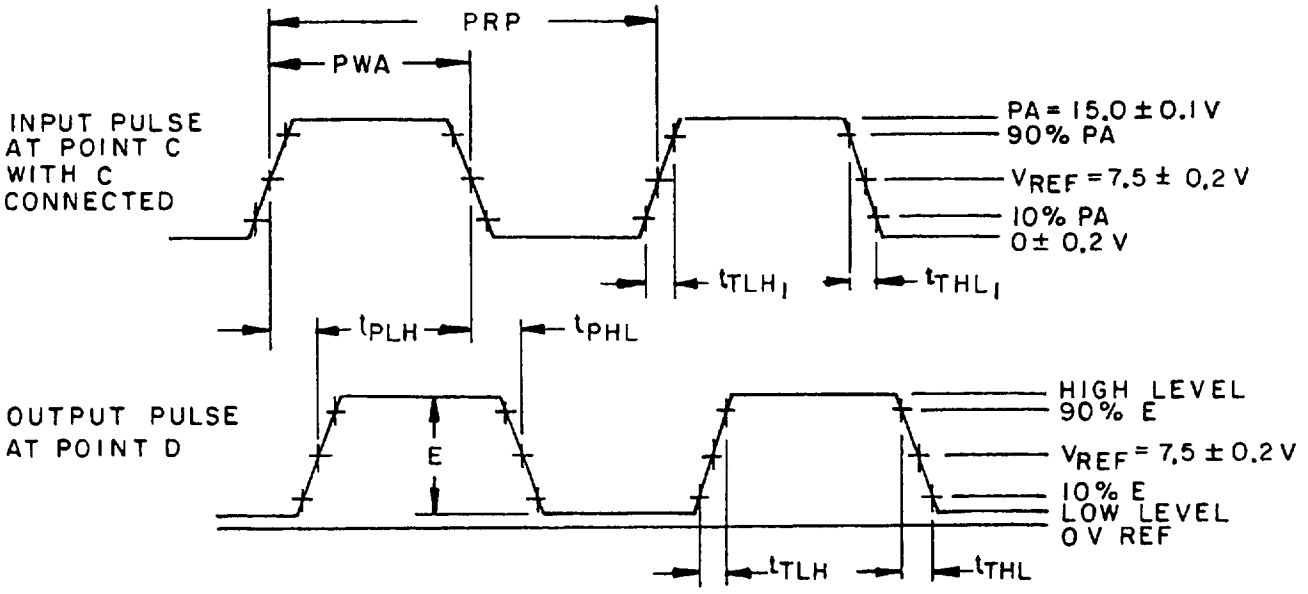
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NOTES
1. Input pulse parameters.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making measurements verify that the output pulse high level is ≥ 13.5 V and the low level is ≤ 1.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1 and YO1-2 to input and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 27 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXXIV through XXXVII.

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NOTES
1. Input pulse parameters
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 1.0 ±0.1 µs.
   PRP = 5.0 ±0.5 µs.
2. Before making measurements verify that the output pulse high level is ≥ 13.5 V and the low level is ≤ 1.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins, probes YO1-1 and YO1-2 to input and output pins shall be less than 1.00 inch (25.4 mm).
4. All voltages are referenced to V_{SS}

FIGURE 28. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XXXVI and XXXVII.

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TABLE XXXVIII. Disable delay time (t_{PZH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
MM1	Disable delay time (t_{PZH}), C_L = 50 pF, V_{DD} = 15 V	Open	23,6,28,31,15,18	26	7	8	80	ns	29,30
MM2				26	22				
MM3				11	30				
MM4				11	29				
MM5				38	17				
MM6				38	16				

TABLE XXXIX. Disable delay time (t_{PHZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
NN1	Disable delay time (t_{PHZ}), C_L = 50 pF, V_{DD} = 15 V	Open	23,6,28,31,15,18	26	7	8	80	ns	29,30
NN2				26	22				
NN3				11	30				
NN4				11	29				
NN5				38	17				
NN6				38	16				

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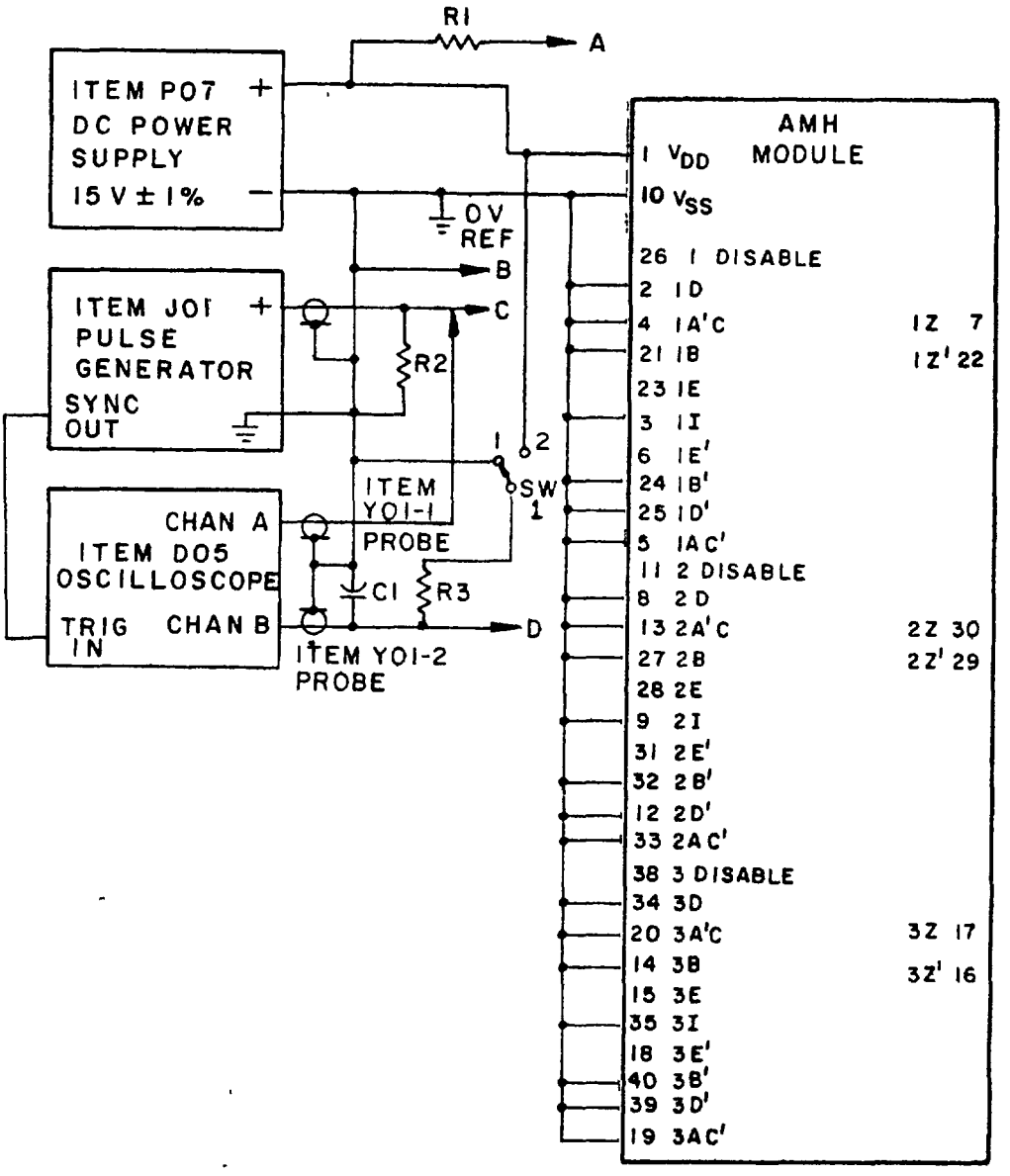
TABLE XL. Disable delay time (t_{PZL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
PP1	Disable delay time (t_{PZL}), C_L = 50 pF, V_{DD} = 15 V	23,6,28,31,15,18	Open	26	7	8	80	ns	29,30
PP2				26	22				
PP3				11	30				
PP4				11	29				
PP5				38	17				
PP6				38	16				

TABLE XLI. Disable delay time (t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
QQ1	Disable delay time (t_{PLZ}), C_L = 50 pF, V_{DD} = 15 V	23,6,28,31,15,18	Open	26	7	8	80	ns	29,30
QQ2				26	22				
QQ3				11	30				
QQ4				11	29				
QQ5				38	17				
QQ6				38	16				

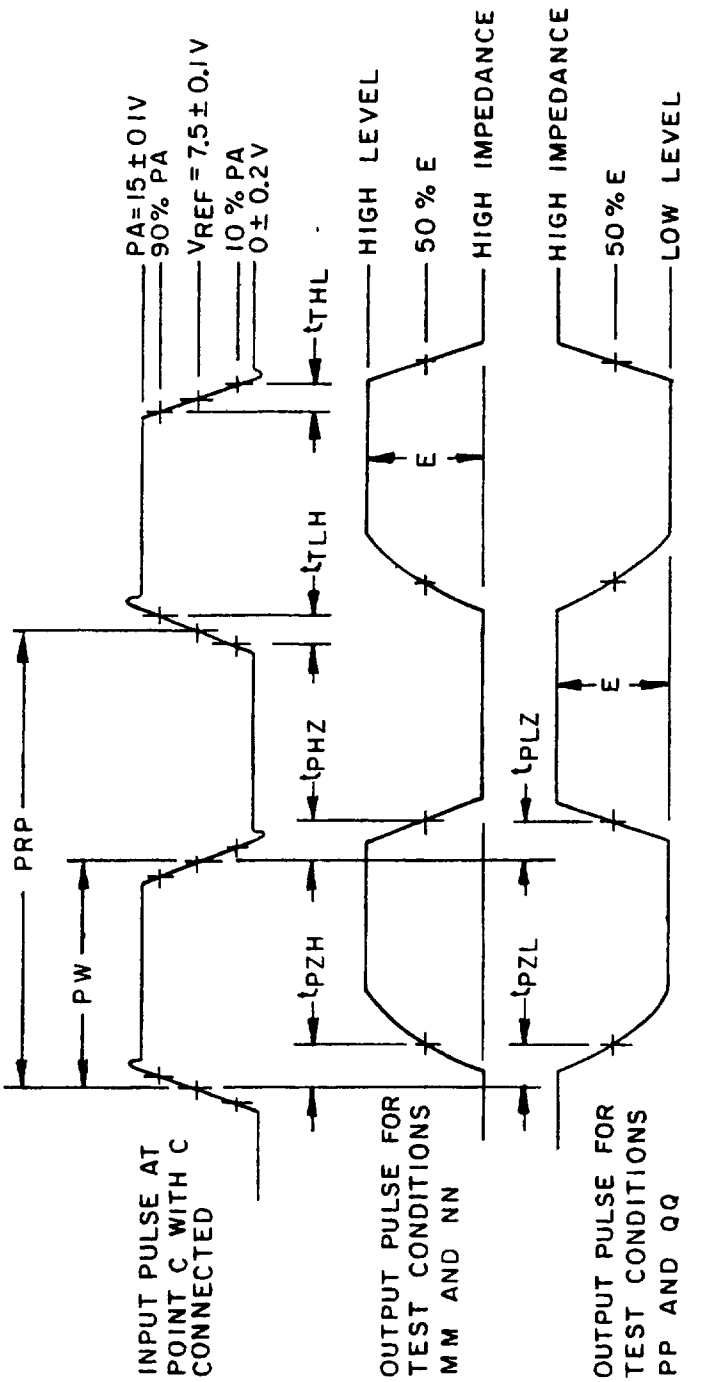
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NOTES:
1. R1 = 1.5 kΩ ±1%, 1/4 W.
2. R2 = 51.1Ω ±5%, 1/2 W.
3. R3 = 1.0 kΩ ±1%, 1/4 W.
4. C1 = 50 pF ±10% (including probe and parasitic capacitance).
5 Place switch SW1 in position 1 for test conditions PP1-PP6 and QQ1-QQ6, and in position 2 for test conditions MM1-MM6 and NN1-NN6.

FIGURE 29. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit for tables XXXVIII through XLI.

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NOTES:
1. Input pulse parameters:
   PW = 8 ±0.5 µs.
   PRP = 16 ±1.0 µs.
   $t_{TLH} = t_{THL}$ = 20 ±2 ns.
2. Before making the measurements specified verify the output pulse high level is ≥ 13.5 V and the low level is ≤ 1.5 V.

FIGURE 30. Disable delay time (t_{PZH}, t_{PHZ}, t_{PZL}, and t_{PLZ}) test circuit waveforms for tables XXXVIII through XLI.

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TABLE XLII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
RH	Power supply filter capacitance	Open	1	0.007	0.013	µF	31

TABLE XLIII. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
SS1	1 DISABLE Input capacitance	26	Open	2	15	pF	31
SS2	1D	2					
SS3	1B	21					
SS4	1E	23					
SS5	1I	3					
SS6	1E'	6					
SS7	1B'	24					
SS8	1D'	25					
SS9	2 DISABLE	11					
SS10	2D	8					
SS11	2B	27					
SS12	2E	28					
SS13	2I	9					
SS14	2E'	31					
SS15	2B'	32					

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TABLE XLIII. Input capacitance test - Continued.

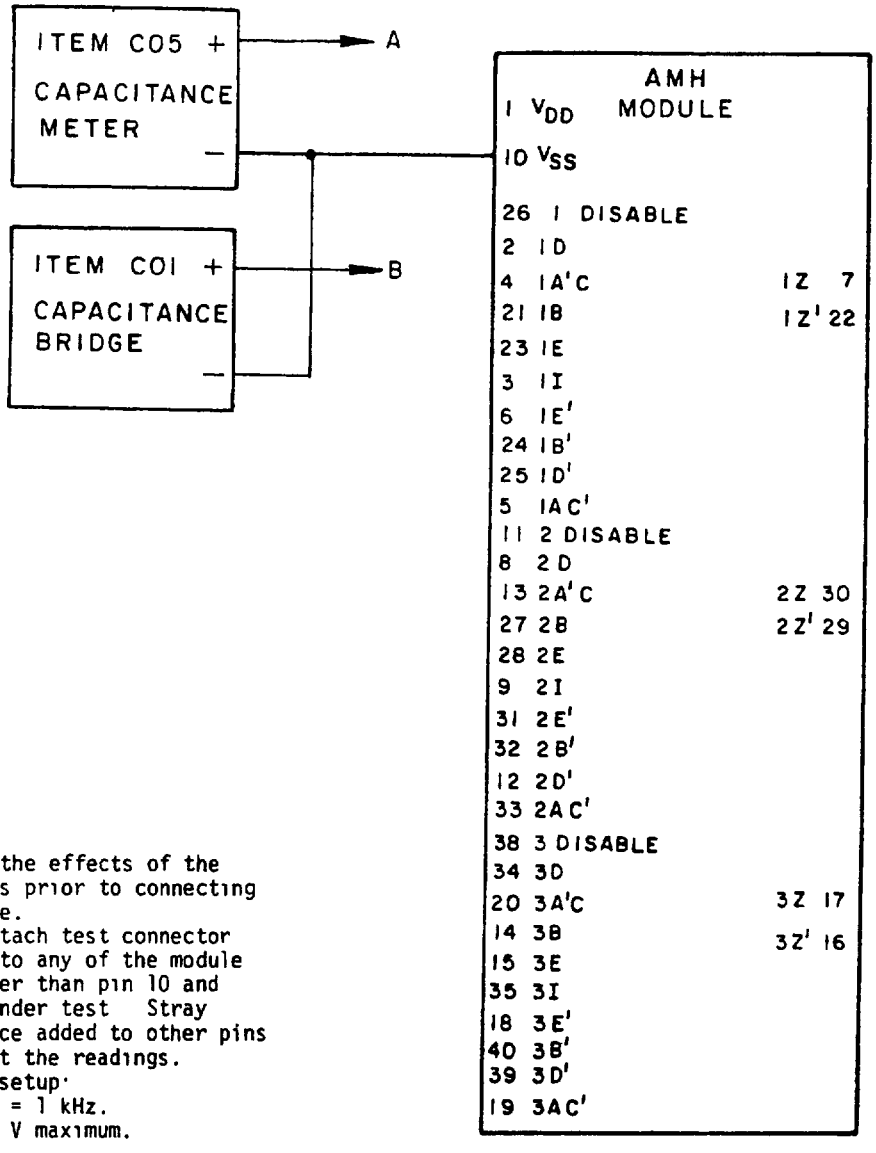
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
SS16	2D' Input capacitance	12	Open	2	15	pF	31
SS17	3 DISABLE	38					
SS18	3D	34					
SS19	3B	14					
SS20	3E	15					
SS21	3I	35					
SS22	3E'	18					
SS23	3B'	40					
SS24	3D'	39					
SS25	1A'C	4		4	25		
SS26	1AC'	5					
SS27	2A'C	13					
SS28	2AC'	33					
SS29	3A'C	20					
SS30	3AC'	19					

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TABLE XLIV. Output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
TT1	1Z	Output capacitance	7	Open	2	15	pF	31
TT2	1Z'	↓	22	↓	↓	↓	↓	↓
TT3	2Z		30					
TT4	2Z'		29					
TT5	3Z		17					
TT6	3Z'		16					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins, other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3 Item C01 setup:
Frequency = 1 kHz.
V_{pk} = 0.3 V maximum.

FIGURE 31 Power supply filter capacitance, input and output capacitance test circuit for tables XLII, XLIII, and XLIV.

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